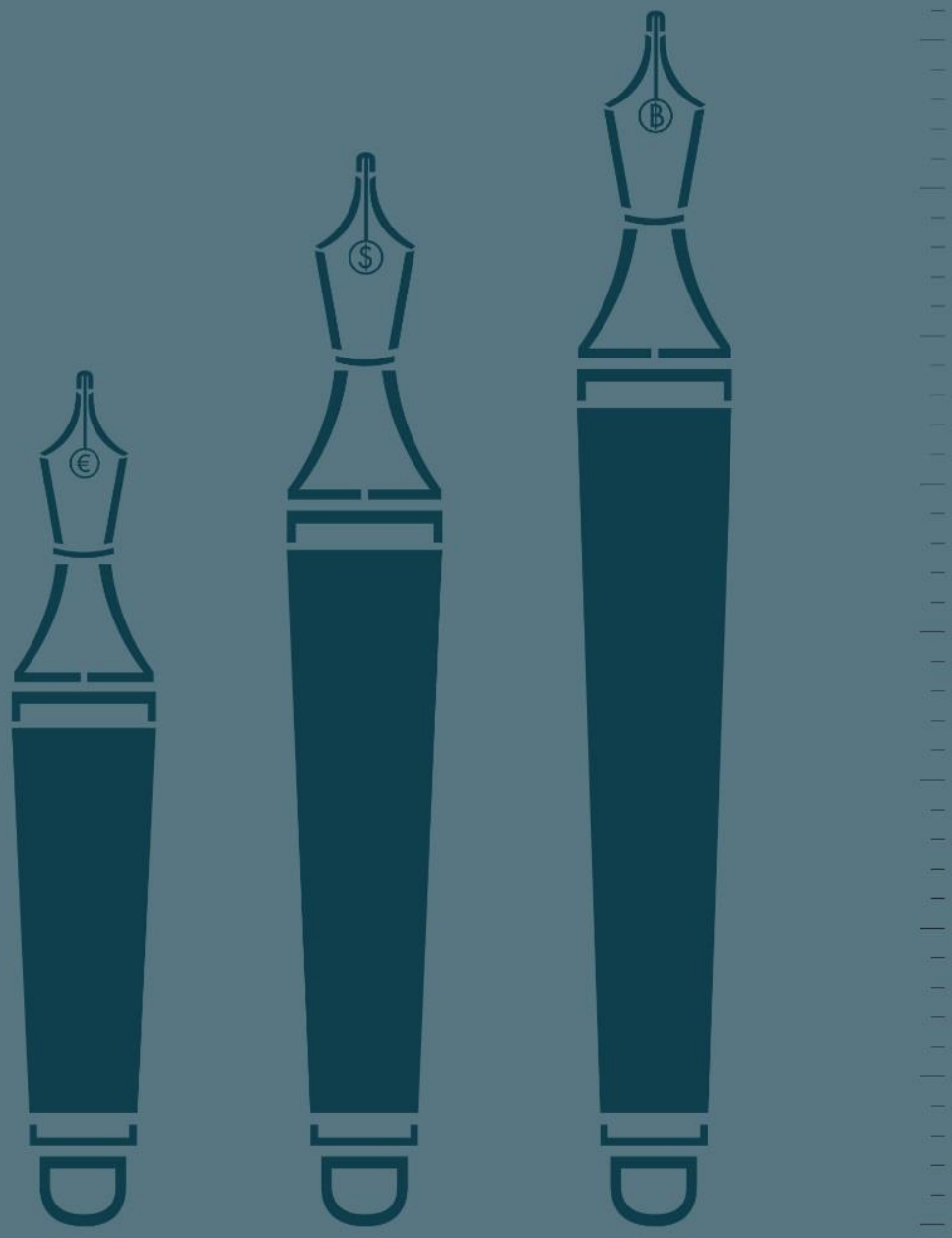
BANK OF THAILAND

MONETARY POLICY REPORT JUNE 2016





BANK OF THAILAND

Monetary Policy Report

June 2016

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
June 2016

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for 2016 and for the medium term. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

Monetary Policy Report

June 2016

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is projected to grow in 2016 at the same rate as estimated in the previous *Monetary Policy Report* at 3.1 percent. Exports remain constrained by both weak economic growth among trading partners and structural changes in global trade. However, the adverse impact is offset by expansion in exports of services, thanks to the increase in the number of foreign tourists, and by private consumption that is supported by government spending and stimulus measures as well as improved income growth among agricultural households following rising agricultural commodity prices. For 2017, growth is slightly revised down from the previous estimate to 3.2 percent on account of a slowdown in merchandise exports, while domestic demand and tourism will continue to drive growth going forward.

Headline and core inflation this year is expected to remain close to the previous assessment. Cost-push inflationary pressure is largely unchanged from the previous projection, as upward adjustments in domestic oil prices are offset by lower electricity prices. Demand-pull inflationary pressure is broadly unchanged, which is in line with the growth prospect that is mostly similar to the previous projection. For 2017, headline inflation is expected to pick up gradually in tandem with the gradual rise in global oil prices. Core inflation is also projected to pick up, albeit at a slightly slower pace compared with the previous assessment, in tandem with the economy that will grow at a slightly more moderate pace than expected.

The Committee kept the policy interest rate unchanged at the May and June 2016 meetings. According to the Committee's assessment, the Thai economy continues to recover despite downside risks on both the domestic and external fronts. Inflation will return to the lower bound of the inflation target within the latter half of this year. Meanwhile, monetary conditions remain accommodative and conducive to the economic recovery, as reflected in the real interest rates and bond yields that remain at low levels. The Committee emphasizes the need to preserve policy space as cushion for future risks that have the potential to negatively affect the ongoing recovery. The Committee will continue to monitor risks to financial stability, particularly the search-for-yield behavior and underpricing of risks that may arise in the prolonged low interest rate environment.

1.1 Growth and inflation prospects

The Committee maintains the GDP growth forecast for 2016 at 3.1 percent (Table 1.1). Key growth drivers consist of (1) higher-than-expected exports of services that expand on the back of strong growth in the tourism sector and (2) continued government expenditure, underpinned by effective disbursement as well as additional stimulus measures, that helps bolster private-sector confidence and spending. The role of government expenditure is critically important especially when merchandise exports have yet to recover, given the sluggish growth in trading partners' economies, especially in Asia, and changes in the global trade structure.

Table 1.1 Forecast summary

Percent	2015*	2016	2017
GDP growth	2.8	3.1	3.2
		(3.1)	(3.3)
Headline inflation	-0.9	0.6	2.2
		(0.6)	(2.2)
Core inflation	1.1	0.8	1.0
		(0.8)	(1.1)

Note: *Outturn
() March 2016 MPR
Sources: Office of National Economic and Social Development Board, Ministry of Commerce; calculations by Bank of Thailand

Private consumption is projected to expand at a rate close to the previous assessment. Tourism receipts and incomes of agricultural households, the latter of which grow faster than expected due to rising agricultural commodity prices, help offset partly the decline in the incomes of households that involve with export-oriented manufacturing. Meanwhile, private investment is projected to improve relative to the previous forecast due mainly to the outlays for construction in connection with government projects. However, outlays for machinery and equipment are expected to remain at low levels consistent with weak merchandise exports.

For 2017, domestic demand will continue to be the main growth driver, although its momentum is estimated to soften slightly from the previous projection. This is a result of (1) subdued merchandise exports held down in part by structural problems in global trade and (2) the waning of the stimulus effects of government measures on private-sector expenditure. **The**

Committee thus revises down the growth projection next year to 3.2 percent, down from 3.3 percent (Table 1.1). This is consistent with the output gap that is projected to widen somewhat in 2017 compared to the previous estimate (Chart 1.1).

Consistent with the growth projection above, the **forecast of core inflation for 2016 is kept unchanged at 0.8 percent, while the 2017 projection is revised down to 1.0 percent from 1.1 percent.** In addition, this year's cost-push inflationary pressure is estimated to be mostly unchanged from the previous projection, since increase in domestic oil prices in tandem with an upward revision of global crude prices is offset by lower electricity prices. **In sum, the Committee keeps unchanged the projection for headline inflation for 2016 at 0.6 percent, consistent with core inflation and cost-push inflation that are broadly unchanged this year, and maintains the headline inflation projection for 2017 at 2.2 percent, as the effect of the rising cost-push inflationary pressure is estimated to be offset by the declining demand-pull inflationary pressure. (Table 1.1).**

The Committee has incorporated key economic developments into the growth and inflation forecasts as summarized below.

(1) Trading partners' economies are likely to grow at a slower pace than previously assessed (Table 1.2).

The Committee revises down the projection of trading partners' GDP growth for 2016, as Asian economies are expected to slow down further than previously anticipated. This is due to both the changes in the global trade structure that will negatively affect intra-regional trade and the slower-than-expected recovery of advanced economies.

Chart 1.1 Output gap

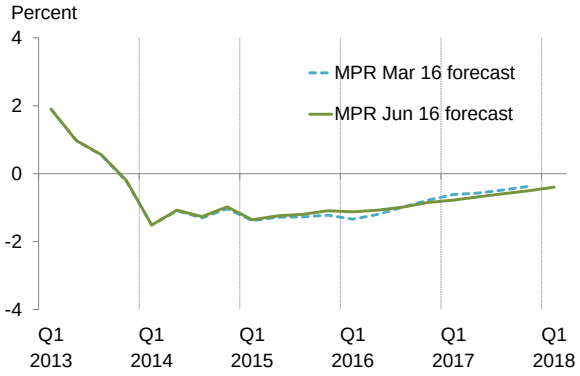


Table 1.2 Growth assumptions for Thailand's trading partners

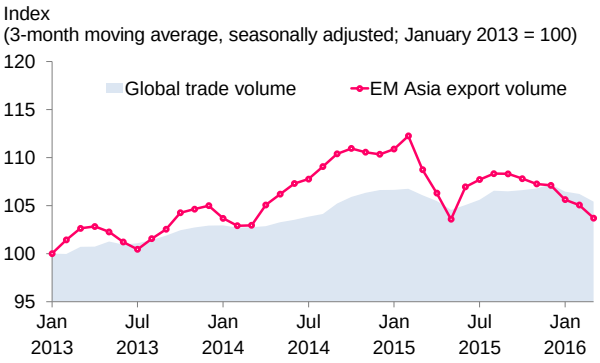
Percent (%YoY)	Weight (%)	2015	2016		2017	
			Mar 16	Jun 16	Mar 16	Jun 16
United States	14.9	2.4	2.2	1.9	2.4	2.3
Euro area	10.0	1.6	1.4	1.7	1.6	1.6
Japan	13.6	0.6	0.8	0.6	0.4	1.0
China	15.7	6.9	6.5	6.5	6.4	6.4
Asia (ex Japan and China)*	37.4	3.5	3.5	3.2	3.9	3.7
Total**	100	3.2	3.1	3.0	3.3	3.3

Note: * Weighted by each trading partner's share of Thailand's total exports in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), Korea (2.8%), and the Phillippines (3.7%)
** Weighted by each trading partner's share of Thailand's total exports as of 2014

For 2017, trading partners' GDP growth projection remains unchanged. While the US and Asian economies are expected to record weaker growth, it is anticipated that Japan will expand faster than previously projected, given that the increase in the sales tax, initially scheduled for 2017, has now been pushed back to 2019. Meanwhile, growth in the euro area and China is expected to slow down in line with the previous assessment.

In addition to weaker growth of trading partners, changes in the global trade structure, whereby import dependencies have reduced in many economies, contribute to the continuous decline in both the global trade volume and Emerging Asia's export volume in particular **(Chart 1.2)**. Such development is one of the key factors that leads to greater contraction in Thailand's merchandise export volume than previously anticipated. Consequently, Thailand's export value is estimated to decline despite an upward revision of export prices thanks to higher-than-expected commodity and crude oil prices. **The Committee revised down the export value projection to a contraction of 2.5 percent in 2016, larger than the 2.0 percent decline according to the previous assessment. The export value for 2017 is projected to remain unchanged from the prior year's level, close to the 0.1 percent growth estimated previously.**

Chart 1.2 Global trade volume and Emerging Asia's export volume



Source: Netherlands Bureau for Economics, calculation by Bank of Thailand

Given the sluggish global economic recovery, monetary policy in major economies is likely to remain accommodative. The Federal Reserve has maintained an accommodative monetary policy stance for longer than the Committee previously expected. Most recently, the federal funds rate was kept on hold at the latest Federal Open Market Committee (FOMC) meeting in June, following the release of May employment

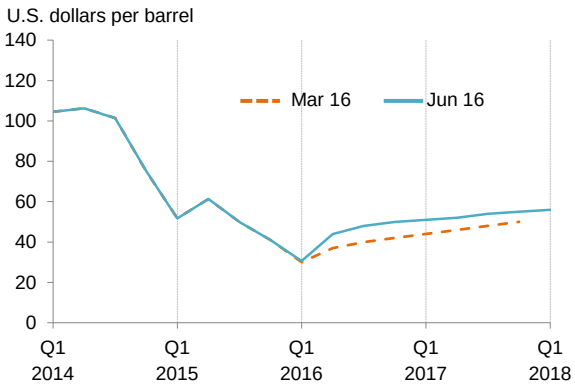
figures that were below market expectations, and also in anticipation of the results of the United Kingdom (UK) European Union (EU) membership referendum that could spark volatility in global financial markets. The Committee expects that the FOMC is likely to wait for clearer signs of recovery in the US labor market before deciding on a rate hike in the latter half of this year and continuing with another rate hike next year. Meanwhile, the Bank of Japan has continued with monetary easing through quantitative and qualitative easing with the negative interest rate policy. Looking ahead, monetary policy divergence among major advanced economies will continue to be an important factor contributing to volatilities in capital flows and exchange rates going forward.

The Committee expects that trading partners' economies will be faced with greater downside risks. One of these risks is China's financial stability concerns which stem from elevated corporate debt that potentially results in financing difficulties for businesses and state-owned enterprises, thereby increasing default risks and costs of funds. Additional risks that warrant close monitoring are (1) uncertainties in monetary policy of major economies and (2) the UK's exit from the EU (Brexit). These risks could add to volatility in global financial markets and could have greater economic impacts than expected.

(2) Crude oil prices in the global market, which were higher than previously expected in the second quarter of 2016, are projected to rise over the forecast horizon (Chart 1.3).

The Committee revises up the projection for Dubai crude price throughout the forecast horizon. The oil price is expected to average at 43.1 US dollars per barrel in 2016, up from the previous estimate of 37.3 US dollars per barrel. This upward

Chart 1.3 Assumptions on Dubai oil price



revision is due to the more-than-expected increase in oil prices in the second quarter of this year as a result of supply-side factors. Oil production in many OPEC countries has slowed, putting upward pressure on prices both recently and going forward. In particular, lingering political tensions in Nigeria and financial problems in Venezuela are major factors behind the supply shortage. Meanwhile, production in non-OPEC countries remains low on account of US producers having withdrawn from exploration and drilling of shale oil. Looking ahead, the Committee thus expects that the global oil market will enter into new equilibrium more quickly, which will prompt US shale oil producers to resume production once prices exceed the break-even point, and that the increase in oil prices is expected to decelerate around the end of the forecast horizon.

The more-than-expected increase in global oil prices will result in higher domestic retail oil prices compared with the previous projection. However, domestic energy prices as a whole are not expected to significantly increase due to the decline in electricity prices[1/] that will offset this year's higher oil prices.

Prices of agricultural products have risen alongside crude oil prices, helping support farm income growth and sustain household consumption. Moreover, higher crude oil and other commodity prices lead to increasing export prices, helping

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[1/] The Energy Regulatory Commission (ERC) approved the decrease of the float time (Ft) surcharge in May through to August 2016 by 28.49 satang per unit. This resolution was based on the cost of natural gas used to fuel power generation that fell over the previous six months, together with the lower electricity purchase costs by the Electricity Generating Authority of Thailand (EGAT) as privately owned electric power plants were unable to deliver electrical currents in accordance with their contractual agreement with EGAT.

cushion somewhat the effects of the decline in the merchandise export volume. Notwithstanding, overall farm income remains low, owing partly to the fact that higher prices still cannot fully offset the impact of the drought. This, together with elevated levels of household debt, will weigh on private consumption going forward.

The Committee judges risks to crude oil prices to be balanced. Downside risks are as follow: (1) global oil demand may be lower than anticipated given a delayed global recovery; (2) shale oil producers may resume production sooner than the baseline projection as oil prices rises; and (3) supply of crude oil from OPEC producers may be larger than expected as they increase production to maintain their market share. On the other hand, upside risks come mainly from conflicts in the Middle East which potentially spread to major oil production locations.

(3) The number of tourists has risen more than expected as a result of continued tourist growth in almost all nationality groups. Going forward, growth in the number of tourists is estimated to exceed the previous projection.

Exports of services are expected to grow strongly on the back of increases in the number of tourists that is higher than previously assessed. Particularly, the number of Chinese tourists has increased significantly, while other nationalities have shown clearer signs of recovery. The number of Russian tourists, in particular, has turned around with positive growth after two consecutive years of contraction as the Russian economy has gradually recovered. Moreover, the expansion of low-cost airline routes, both of local and international airlines, has been a key supporting factor to the robust growth in Thailand's tourism sector. The Committee estimates the number of foreign tourists to rise to 34 million in

2016, up from the previous assessment of 32.4 million, and 36.7 million in 2017, up from the previous projection of 34.4 million.

Improved employment in the service sector thanks to tourism has partly offset income loss in the manufacturing sector following the export slowdown. The strength of the service sector thus contributes to an overall expansion in household income which has supported private consumption over the recent period. However, going forward, the service sector may not be able to absorb labor reallocation from other sectors, as labor demand in the service sector has started to soften. Moreover, elevated household debt has induced more cautious spending behavior. In this regard, private consumption is expected to recover at a slow pace.

(4) The public sector continues to play a critical role in sustaining the economy through public expenditure, which is consistent with the previous assessment, as well as through additional stimulus measures.

Overall, the public sector's disbursement for investment spending and financial commitments is continuously moving forward. The Committee thus revises up the projection for public consumption following the expedited disbursement of transfer payments to the National Health Security Fund in the first quarter of this year. Meanwhile, the Committee expects public investment in 2016 to remain close to the previous projection and to rise slightly in 2017 due to expedited carry-over expenditure in accordance with the Transfer of Expenses Budget Act, B.E. 2559. The outlook for investment of state-owned enterprises (SOEs) also remains largely unchanged despite some delay in the Prachuap Khiri Khan – Chumphon dual-track rail project.

Public expenditure has been well on-track, especially those related to government measures to

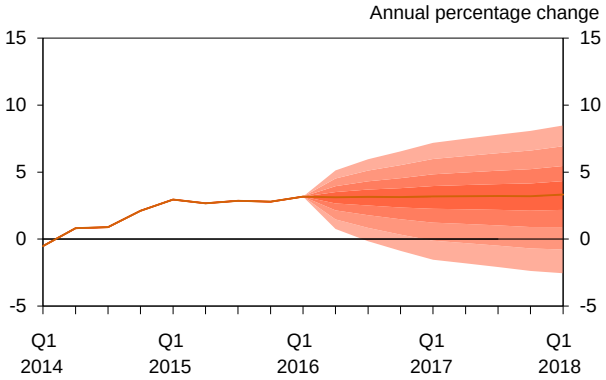
improve living standards in provincial sub-districts. These measures involve transfers to sub-district agencies for the purpose of consumption and investment, which in effect directly induces private-sector spending. In addition, the public sector's outlays for construction have helped accommodate labor movement out of the agricultural sector over the recent period. This, in combination with the recovery in domestic demand, will in the short run aid private investment to grow faster than the previous assessment. Nonetheless, private investment is expected to moderate going forward due to the waning of effects from stimulus spending and due to more subdued external demand that potentially affect future earnings and investment plans.

The Committee judges the balance of risks to public spending to be tilted to the upside. Upside risks include (1) some SOE investment projects—including additional dual-track rails, high-speed rail projects, and new express-way routes—that may commence next year, earlier than previously expected, and (2) higher-than-expected expenditure by local governments thanks to measures to expedite disbursement of the accumulated funds of local administrative organizations. Downside risks include (1) limited efficiency of spending disbursement by some government agencies, (2) rescission of funds already made available but not yet entered into financial commitments, and (3) political uncertainties which can pose a delay in public expenditure.

Risks to Growth and Inflation Forecasts

The Committee views risks to growth to remain tilted to the downside, as reflected in the growth fan chart that is skewed downward throughout the forecast horizon (Chart 1.4). Downside risks to growth are (1) weaker-than-expected growth of trading partners, particularly China as a result of financial stability concerns, (2)

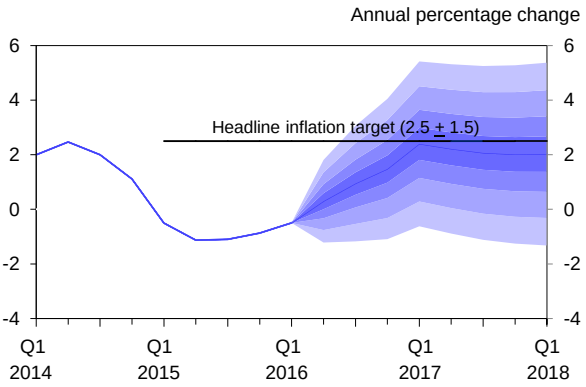
Chart 1.4 GDP growth forecast



Note: Fan chart covers 90 percent of probability distribution

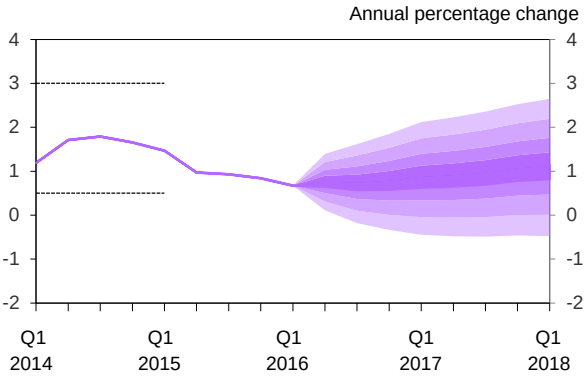
households that are financially impaired and have less financial cushion for economic uncertainties than previously expected, and (3) political uncertainties that may affect public spending and tourism. Nevertheless, upside risks include (1) government investment that may be undertaken sooner and greater than expected, together with the effects of government stimulus that may turn out greater than the baseline projection, and (2) the number of tourists that may be greater than the baseline projection. With regard to inflation, the Committee judges **the balance of risks to both headline and core inflation to be tilted to the downside in line with the balance of risks to growth (Chart 1.5 and 1.6).**

Chart 1.5 Headline inflaltion forecast



Note: Fan chart covers 90 percent of probability distribution

Chart 1.6 Core inflation forecast



Note: Fan chart covers 90 percent of probability distribution

1.2 Monetary policy decision

Monetary policy remained accommodative in the second quarter of 2016 to support the ongoing economic recovery that continues to face downside risks on both domestic and external fronts. The Committee will continue to closely monitor financial stability risks arising from prolonged low interest rates and emphasizes the need to preserve policy space given remaining uncertainties over the economic outlook.

At the MPC meeting on May 11, 2016, the Committee voted unanimously to maintain the policy rate at 1.50 percent per annum. In deliberating this decision, the Committee assessed that **the Thai economy continued to expand at a pace close to the assessment in the previous meeting. Headline inflation was expected to pick up gradually after turning positive in April,** as the base effect of oil price dissipated over time, although demand-pull inflationary pressure still

remained subdued overall. The balance of risks to economic growth was judged to be tilted more to the downside as the impact of the drought on agricultural households was likely to weigh on private consumption going forward.

Meanwhile, monetary conditions eased further because of lower bond yields and the recent cuts in lending rates by commercial banks. Moreover, monetary conditions were judged to remain accommodative despite the fact that the Thai baht appreciated against some trading partners' currencies at some points which might have not been as conducive to the economic recovery as it could have been.

The Committee viewed that keeping the policy rate on hold at this meeting would help preserve policy space for risks that could have negative impact on the ongoing economic recovery. These risks included a possible delay in the global economic recovery, weaker consumption growth, and political uncertainties that might affect both private investment and consumption. Additionally, some committee members viewed the current economic slowdown to be attributed to structural problems, both domestically and globally, which could not be addressed by monetary policy alone. Moreover, cutting the policy rate at this juncture could potentially lead to further accumulation of imbalances in the financial sector, especially given that interest rates remained low for an extended period. The Committee would also continue to closely monitor risks to financial stability, particularly the search-for-yield behavior that could exacerbate underpricing of risks, and the possibility of a snapback of global bond yields, which could affect financing conditions, balance sheets of businesses, and the overall monetary conditions through adjustments in Thai government bond yields.

In the following MPC meeting on June 22, 2016, the Committee voted unanimously to maintain the policy rate at 1.50 percent per annum. **The Committee projected that the Thai economy would continue to recover and inflation would return to the target band in the latter half of the year in line with previous assessment.** Meanwhile, monetary conditions remained accommodative and conducive to the ongoing recovery. Looking ahead, the Committee viewed that monetary policy should continue to be accommodative and would continue to closely monitor financial stability risks.

In deliberating this decision, the Committee gave due consideration to **Thailand's ongoing economic recovery**. Going forward, the economy would be driven primarily by public expenditure and tourism. At the same time, private consumption would improve in line with previous assessment while private investment would continue to be at a low level. However, merchandise exports were projected to contract in tandem with slower-than-expected growth of Asian economies. In light of these developments, the Committee assessed that the economic momentum from domestic demand and tourism would help offset the impact of weaker exports. As a result, the economic outlook would be largely unchanged and the economy would expand at the same pace as assessed in the previous meeting. Nevertheless, there remained several downside risks to growth, namely weaker-than-expected growth in trading partners and fragility in private-sector sentiments. Incidentally, previous concerns pertaining to the impact of the drought subsided along with the recovery in prices of some agricultural commodities.

The Committee also gave due consideration to **the inflation outlook where headline inflation was projected to return to the lower bound of the inflation target within the latter half of 2016**

as previously expected. Meanwhile, medium-term inflation expectations remained well-anchored around the inflation target. Demand-pull inflationary pressure increased somewhat due to the ongoing economic recovery, as reflected in core inflation that edged up slightly. Headline inflation was projected to pick up gradually as energy prices would rise in tandem with crude oil prices. Medium-term inflation expectations (5-10 years ahead) based on a survey of professional economists' remained well-anchored around the inflation target of 2.5 percent.

Monetary conditions were accommodative throughout recent periods and were conducive to the economic recovery and the return of headline inflation to the target. Accommodative monetary conditions were reflected in low real interest rates and the continued expansion of total corporate financing and lending to households, although certain businesses faced limitations in accessing credits. Going forward, the Committee still saw the need to continue monitoring financial stability risks, including the search-for-yield behavior, in the prolonged low interest rate environment.

Finally, the Committee assessed the **preservation of policy space to be critical**, given that the economy would face various risks going forward. These risks included the fragile global economic recovery, monetary policy divergence, financial stability concerns in China, as well as Brexit. The Committee would stand ready to utilize an appropriate mix of available policy tools in order to make sure that monetary conditions and the exchange rate remain conducive to the economic recovery as well as to ensure financial stability.

1.3 Appendix: Tables for supporting assumptions and forecasts

Table 1.3 Forecast for GDP and assumptions			
Percent	2015*	2016	2017
GDP growth	2.8	3.1	3.2
Domestic demand	2.8	3.0	2.5
Private consumption	2.1	1.8	2.1
Private investment	-2.0	3.1	2.3
Government consumption	2.2	3.5	2.8
Public investment	29.8	10.1	5.2
Exports of goods and services	0.2	2.2	0.9
Imports of goods and services	-0.4	-1.9	2.4
Current account (billion, US dollars)	32.0	37.8	32.3
Value of merchandise exports	-5.6	-2.5	0.0
Value of merchandise imports	-11.3	-6.0	5.3

Note: *Outturns

Table 1.4 Forecast assumptions			
Annual percentage change	2015*	2016	2017
Dubai oil price (U.S. dollars per barrel)	50.8	43.1	53.0
Non-fuel commodity prices (%YoY)	-17.5	-5.5	2.6
Fresh food prices (%YoY)	-4.4	3.0	3.7
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY)	4.4	6.3	6.3
Public investment (current price) (%YoY)[1/]	25.6	10.4	8.4
Fed Funds rate (% at year-end)	0.38	0.63	1.38
Trading partners' economic growth (% YoY)[2/]	3.2	3.0	3.3
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	150.7	154.4	157.2

Note: [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by the minus sign
* Outturns

Table 1.5 GDP growth forecasts by research houses		
	2016	2017
Maybank Kim Eng	4.0	5.0
PFO[1/]	3.3	-
NESDB[2/]	3.3	-
Thanachart Securities	3.3	4.2
Bank of Ayudhya	3.2	-
BOT	3.1	3.2
Moody's	3.0	3.4
ANZ Bank	3.0	3.2
Kasikorn Research	3.0	3.2
HSBC	3.0	3.1
JPMorgan	2.9	3.0
TMB Bank	2.8	3.3
Siam Commercial Bank	2.8	3.3
Phatra Securities	2.8	3.2
Kiatnakin Bank	2.8	3.2
Nomura	2.7	2.7

Note: Compiled and published by Reuters on June 20, 2016, except:
[1/] Published on April 28, 2016
[2/] Published on May 16, 2016 with the release of GDP data for 2016 Q1
Presented in descending order of 2016 forecasts

Table 1.6 Headline inflation forecasts by research houses		
	2016	2017
Maybank Kim Eng	2.0	2.5
Thanachart Securities	0.8	2.0
BOT	0.6	2.2
ANZ Bank	0.6	2.0
Kasikorn Research	0.6	1.7
Nomura Co Ltd	0.6	1.4
Bank Ayudhya	0.6	-
TMB Bank	0.5	2.5
Kiatnakin Bank	0.5	1.8
Phatra Securities	0.5	1.8
Siam Commercial Bank	0.4	2.3
TISCO Securities	0.4	2.2
JPMorgan	0.4	2.1
NESDB[1/]	0.4	-
FPO[2/]	0.3	-
KT ZMICO	0.0	1.5
Moody's	-0.2	1.7

Note: Compiled and published by Reuters on June 20, 2016, except:
[1/] Published on May 16, 2016 with the release of GDP data for 2016 Q1
[2/] Published on April 28, 2016
Presented in descending order of 2016 forecasts

Table 1.7 Probability distribution of GDP growth forecast									
Percent	2016				2017				2018
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q4
10-12	0	0	0	0	0	0	1	1	2
8-10	0	0	0	1	2	3	3	4	5
6-8	0	1	4	7	9	10	11	11	12
4-6	0	21	24	23	22	22	21	21	20
2-4	100	54	41	34	29	27	26	25	23
0-2	0	22	24	24	22	21	21	20	19
(-2)-0	0	2	6	9	11	11	11	12	12
< (-2)	0	0	1	2	4	5	6	7	7

Table 1.8 Probability distribution of headline inflation forecast									
Percent	2016				2017				2018
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 7	0	0	0	0	0	0	0	1	1
6-7	0	0	0	0	1	1	1	1	1
5-6	0	0	0	1	5	4	4	4	4
4-5	0	0	1	3	11	10	9	9	9
3-4	0	0	4	10	18	17	15	15	15
2-3	0	2	14	20	22	21	20	20	19
1-2	0	18	28	26	20	20	20	20	19
0-1	0	41	29	21	13	14	15	15	15
(-1)-(0)	100	30	17	12	7	8	9	9	9
(-2)-(-1)	0	8	6	5	3	3	4	5	5
(-3)-(-2)	0	1	1	1	1	1	1	2	2
(-4)-(-3)	0	0	0	0	0	0	0	1	1
< (-4)	0	0	0	0	0	0	0	0	0

Table 1.9 Probability distribution of core inflation forecast									
Percent	2016				2017				2018
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 4.5	0	0	0	0	0	0	0	0	1
4.0-4.5	0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	1	1	1
2.5-3.0	0	0	0	0	1	2	2	3	4
2.0-2.5	0	0	1	2	5	6	7	9	10
1.5-2.0	0	2	6	9	13	14	15	16	16
1.0-1.5	0	23	22	23	22	22	21	21	20
0.5-1.0	100	47	35	29	25	23	22	20	19
0.0-0.5	0	24	25	22	19	18	17	15	14
(-1)-0.0	0	3	9	11	10	10	10	9	8
(-2)-(-1)	0	0	2	3	4	4	4	4	4
< -2	0	0	0	1	1	1	1	1	1

Recent Economic Developments

2. Recent Economic Developments

The global economic recovery slowed on account of Asian economies, whose growth weakened as a result of (1) cyclical factors such as the slowdown in China, while the recovery in G3 economies were not sufficient to spur growth in Asian exports and (2) structural factors, including changes in the global trade structure that affected supply chains, a loss of competitiveness, and the transition toward an aging society, that weighed on each economy differently. On the other hand, advanced economies continued to recover gradually, driven by consumption due to improvements in labor market conditions.

The Thai economy continued to recover at a gradual pace on the back of tourism and public spending. Private consumption expanded on account of growth in income and employment in the service sector, but continued to be restrained by the decline in farm income because of the drought and by elevated household debt. Meanwhile, merchandise exports, excluding gold, contracted in tandem with the slowdown among Asian economies, which were Thailand's key trading partners. Such developments caused overall production and investment remaining at low levels.

Headline inflation turned positive with the acceleration in fresh food and energy prices, and core inflation edged up slightly due to higher input costs and demand-pull inflationary pressure that increased somewhat. Going forward, headline inflation was expected to gradually pick up alongside global crude oil prices.

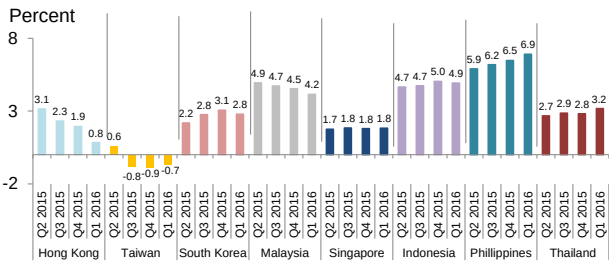
2.1 The global economy

Asia (excluding China and Japan)

The recovery of Asian economies moderated due to exports being weaker than expected as a result of both the slowdown in China and changes in the global trade structure. Meanwhile, advanced economies continued to recover at a gradual pace, although this was not sufficient to spur growth in Asian merchandise exports values.

Asian economies, excluding China and Japan, slowed down in the first quarter of 2016 **(Chart 2.1)** on account of contraction in merchandise

Chart 2.1 GDP growth of Asian economies
(change from same quarter last year)

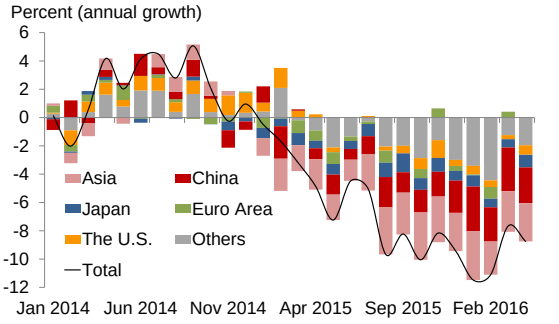


Source: CEIC

exports as a result of weak external demand, especially from China. Meanwhile, the recovery in advanced economies was not able to induce growth in Asian exports **(Chart 2.2)**. Thus, the subdued export outlook continued to weigh down on investment and production in many regional economies. In addition, heightened global financial market volatility during the first quarter led to a broad-based decline in the equity indices of many countries, which negatively affected households' wealth through lower asset prices. Combined with already weak consumer confidence, such developments amounted to a slowdown in private consumption.

Looking ahead, the recovery of Asian economies, excluding China and Japan, was expected to moderate further from the previous assessment in March. This was consistent with an absence of any clear signs of export recovery due to the sluggish global growth. In addition, exports were restrained by structural limitations such as aging population that limited potential growth, loss of competitiveness, and changes in the global trade structure as China and advanced economies turned to domestic-based supply chains. Against this backdrop, the share of trade in world GDP trended downward **(Chart 2.3)**, and therefore Asia's trade would not be able to fully reap the benefits from the recovery of advanced economies. These limitations would continue to drag on Asian exports and economic growth in the period ahead despite the improvement in the terms of trade due to rising oil prices.

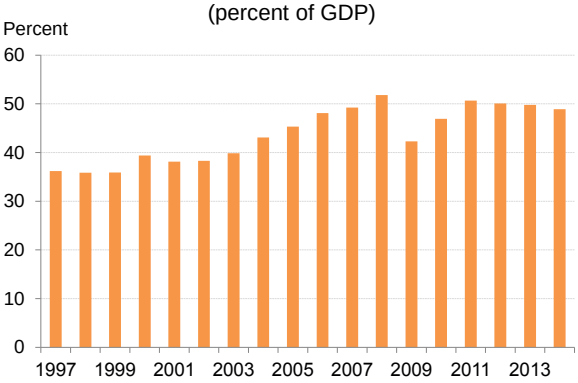
Chart 2.2 Contribution to Asia exports classified by destination



Note: Asia includes Hong Kong, Taiwan, South Korea, Malaysia, Singapore, the Phillippines, and Thailand

Source: CEIC

Chart 2.3 Share of global trade to world GDP (percent of GDP)



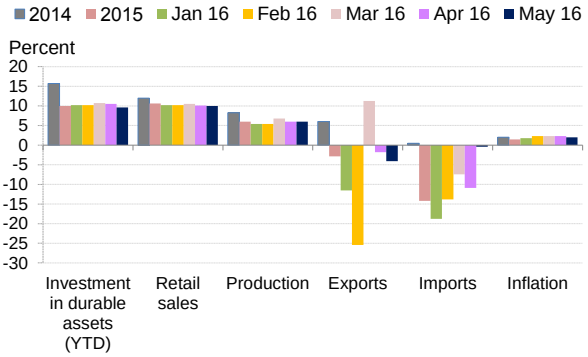
Source: CEIC (World Bank)

China

China's economic growth continued to moderate due to both export contraction and domestic demand, particularly investment, which slowed down as result of ongoing reforms to achieve longer-term economic sustainability.

The Chinese economy expanded at a rate of 6.7 percent (yoy) in the first quarter of 2016, a slight moderation from 6.8 percent in the previous quarter. The marginally lower growth was due to contraction in net exports that stemmed from weak trading partners' demand and investment in the industrial sector which slowed down despite improvements in infrastructure and real-estate investment following stimulus measures to boost short-term growth. Meanwhile, consumption continued to expand. Latest indicators as of May 2016 **(Chart 2.4)** pointed to a continued moderation in China's economic growth, partly because of excess production capacity in many industries which suggested a weak outlook for manufacturing production and investment.

Chart 2.4 China's economic indicators
(percent change from same month last year)



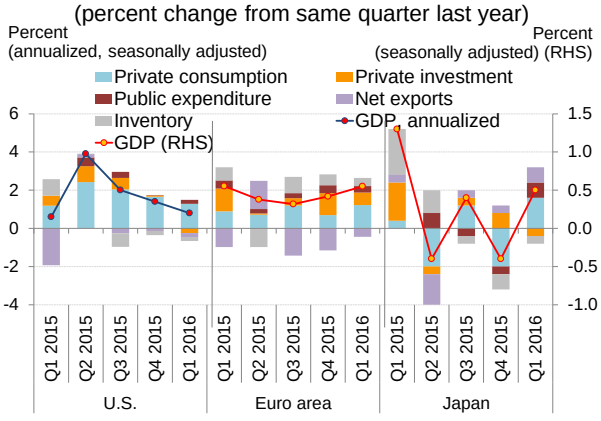
Source: CEIC, Bloomberg

Going forward, growth would slow down further as a result of ongoing economic reforms. China's financial stability continued to warrant close monitoring given heightened default risks in the banking sector and bond markets as well as the shadow banking sector. Consequently, financial conditions in China tightened. This is because of stricter credit standards following signs of weakened financial health in the corporate sector and also because of the greater emphasis by the Chinese authorities on financial stability regulations as one of this year key's economic policies. In any case, China's economic growth this year would likely be within the authorities' target of 6.5–7.0 percent.

Advanced Economies

Advanced economies recovered on the back of domestic demand, especially private consumption that gained momentum on the back of gradual improvements in labor market conditions **(Chart 2.5).**

Chart 2.5 Sources of GDP growth of G3 economies
(percent change from same quarter last year)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

The US economy continued to recover in the first quarter of 2016, albeit at a slower pace of 0.8 percent (qoq saar), down from 1.4

percent in the previous quarter. The slowdown was mainly attributed to the decline in private consumption. This owed partly to the temporal decline in automobile sales following an acceleration in the second half of last year. Meanwhile, private investment contracted, especially in mining and energy sectors which was affected by persistently low oil prices. Investment in residential property, on the other hand, improved in tandem with the recovery in the labor market. In addition, net exports were held down by weak trading partners' demand. Inventories continued to decline.

Latest indicators for the US economy pointed to a continued recovery driven primarily by domestic demand, particularly from private consumption. Retail sales picked up across all product categories, and manufacturing production looked to improve gradually, as reflected in the purchasing manager's index that registered above 50. However, exports remained weak in line with sluggish global growth. In any case, the Federal Reserve would likely delay further hikes in the federal funds rate given the release of the US non-farm payroll data for May 2016 that turned out much weaker than expected.

The euro area economies grew 0.6 percent (qoq sa) in the first quarter of 2016, up from 0.4 percent in the previous quarter, driven by stronger domestic demand, especially in France and Germany where private consumption picked up significantly. These developments were supported by gradual improvements in employment conditions, low oil prices that boosted purchasing power of the private sector, and accommodative monetary policy that benefited the financial sector and helped support lending to both businesses and households. Nevertheless, the decline in exports due to weak global demand continued to weigh on growth.

Going forward, although the euro area would retain forward momentum from domestic demand, fiscal policy would likely be tightened as many member countries were unable to reduce fiscal deficits in accordance with the their budgetary targets. In addition, rising oil prices could pose limitations to growth. Geopolitical risks and political uncertainties in many countries could also affect the economic recovery in the euro area and the global economy. The Committee would therefore closely monitor these risks as they could have implications for global financial markets, Thailand's trading partners' growth, and appropriate monetary policy going forward.

The Japanese economy recorded a growth of 0.5 percent (qoq sa) in the first quarter of 2016, a turnaround from a contraction of 0.4 percent in the previous quarter thanks to expansion in private consumption. Growth in consumer spending was due partly to a rebound from the previous quarter when it slowed down due to the unusually warm weather conditions. Meanwhile, total exports improved on account of exports to the euro area. However, private investment contracted, following two consecutive quarters of growth, as business confidence declined owing to a weaker growth outlook in trading partners, the strengthening of the yen, and heightened financial market volatility at the beginning of the year.

In the period ahead, Japan's economic recovery would likely proceed at a gradual pace. The economy would likely gain further traction thanks to the announced postponement of the increase in the sales tax, initially scheduled for April 2017, for another two and a half years to help sustain consumption. Support would also likely come from additional easing of both monetary and

fiscal policies. Nevertheless, there remained important risks facing the Japanese economy, namely global financial market volatility that could affect domestic demand through more subdued private sector confidence, a potential yen appreciation, and monetary policy transmission to the real economy which might not be as effective given increasing risks on the external front.

2.2 The domestic economy

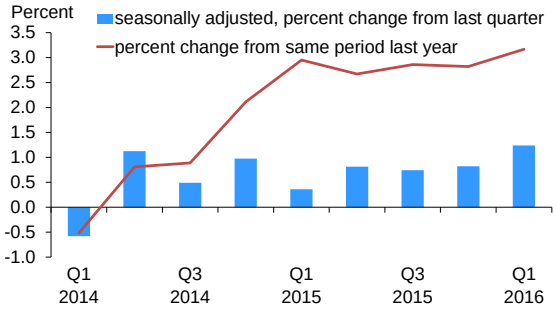
The Thai economy continued to expand at a gradual pace on the back of tourism and public spending. Private consumption was well maintained thanks to growth in income and employment in the service sector. Nevertheless, exports continued to contract in tandem with the slowdown of major trading partners' economies. Consequently, investment remained at a low level **(Chart 2.6)**.

Tourism and public spending still played a crucial role in sustaining economic momentum.

Tourism continued to expand strongly, in part because of Thailand's popularity as a travel destination among Chinese tourists with increasing preferences to travel abroad. In addition, an expansion of low-cost airline routes to Thailand helped boost tourism potential, while the number of visitors from various regions including Asia and Europe showed clearer signs of rebound **(Chart 2.7).** These positive developments therefore benefited the related service industries such as the hotel and restaurant, transportation, and wholesale-retail sectors.

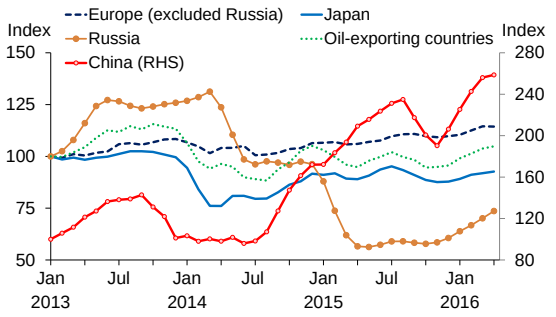
Public spending, especially investment expenditure by the central government, **continued to underpin growth,** despite a slight deceleration

Chart 2.6 GDP growth[1/]
(seasonally adjusted, percent change from last quarter)



Note: [1/] Calculation based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board

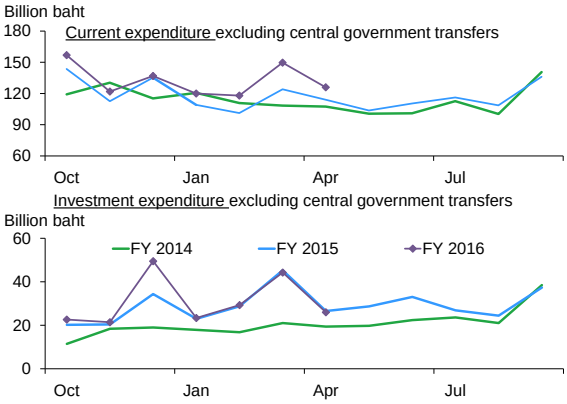
Chart 2.7 Index of foreign tourists classified by nationality
(3-month moving average, seasonally adjusted; January 2013=100)



Note: Oil-exporting countries are Australia, Russia, Indonesia and Malaysia which account for 20% of total foreign tourists
Source: Department of Tourism

following the expedited disbursement at the beginning of the second quarter this year. Meanwhile, capital outlays by state-owned enterprises expanded continuously, sustained in part by outlays for major investment projects, such as the Chira–Khon Kaen and Chachoengsao–Kaeng Khoi dual-track rail projects, that were part of the government's infrastructure development plan. Current spending also expanded well thanks to the government's measures to improve living standards in provincial sub-districts as well as expedited disbursement of funds by the National Health Security Fund in the first quarter **(Chart 2.8)**.

Chart 2.8 Public spending

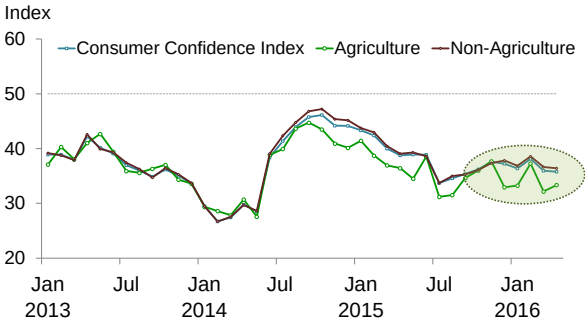


Source: Bureau of Budget; Fiscal Policy Office

Private consumption continued to expand, supported by growth in income and employment in the service sector.

Private consumption continued to expand on account of spending on non-durable goods and services, especially hotel and restaurant services as well as recreation. Nonetheless, automobile purchases still contracted, having accelerated toward the end of last year prior to an increase in the vehicle excise tax. Meanwhile, low purchasing power of agricultural households owing to the drought and elevated household debt remained headwinds affecting consumer confidence and private consumption **(Chart 2.9)** (*Box: The economic impacts of drought on agriculture at the regional level)*. However, sustained growth in income and employment in the service sector could somewhat help sustain overall household purchasing power **(Chart 2.10)**.

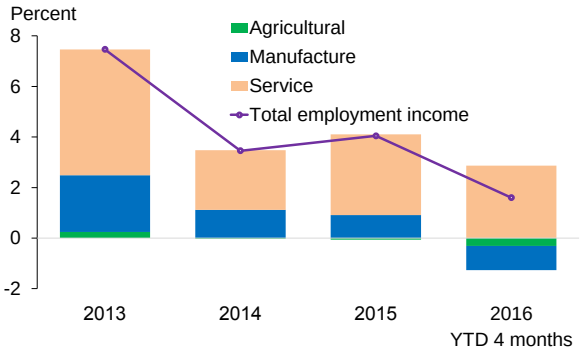
Chart 2.9 Consumer Confidence Index



Note: Share of survey respondents are 76.2% from non-agricultural sector, 11.1% from agricultural sector, and 12.7% are unemployed/studying.

Source: Ministry of Commerce

Chart 2.10 Contribution to employment income growth
(Percent change from last quarter)



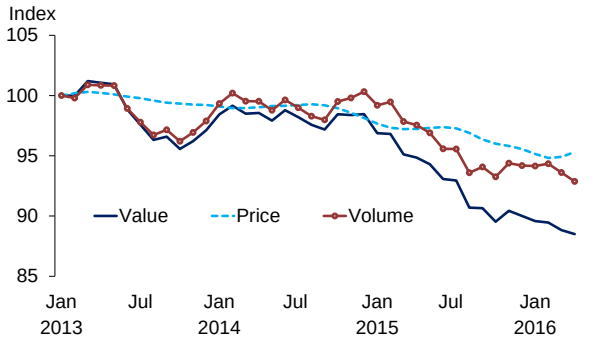
Source: National Statistical Office and Calculations by Bank of Thailand

The value of merchandise exports continued to contract despite an improvement in export prices on the back of rising oil prices.

Global economic slowdown and shifts in the global trade structure continued to adversely affect the export sector in several countries, especially in Emerging Asia which constituted Thailand's most important trading partners. As a result, Thailand's merchandise exports to Asia continued to contract in most product categories, such as exports of machinery and equipment which declined in almost all destination markets. This was especially the case for Asian countries where investment was restrained by export slowdown. Furthermore, Thai exports of petroleum as well as petrochemical and chemical products continued to decline owing to sharp deceleration in demand after China could increase production by itself.

Export prices slightly rebounded from the levels seen at the end of 2015 on the back of incrasing global crude oil prices (Chart 2.11). Oil prices increased after the supply by the Organization of Petroleum Exporting Countries (OPEC) trended lower following a decline in oil production in a number of countries, including Nigeria and Venezuela.

Chart 2.11 Thai exports (excluding gold) value, price, and volume indices
(3-month moving average, seasonally adjusted; January 2013 = 100)

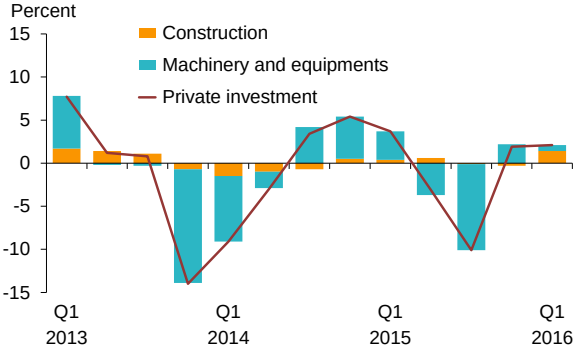


Source: Customs Department, Ministry of Commerce calculations by Bank of Thailand

Export contraction and slow recovery of domestic demand caused growth in private investment to remain low.

Private investment growth remained low. Despite an acceleration in the first quarter of this year due to construction outlays underpinned by government investment stimulus measures, in particular measures to improve living standards in provincial sub-districts, outlays for machinery and equipment remained low **(Chart 2.12)**. Investment in the manufacturing sector continued to be

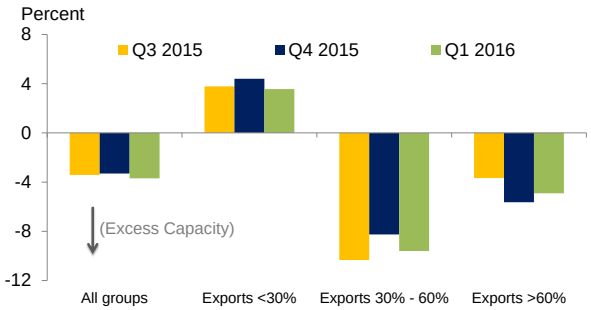
Chart 2.12 Contribution to private investment growth
(change from the same period the previous year)



Source: Office of National Economic and Social Development Board

restrained by contraction in the merchandise export sector, weak domestic business sentiment, and excess production capacity—all of these made it unnecessary to further invest **(Chart 2.13)**. Nonetheless, investment in the service sector continued to expand, although growth was low and only limited to some businesses such as telecommunications and alternative energy.

Chart 2.13 Capacity utilization



Note: Calculated from the difference between capacity utilization in each quarter and 2011-2013 average, excluding period of severe flooding (October 2011 – January 2012)

Source: Calculations by Bank of Thailand

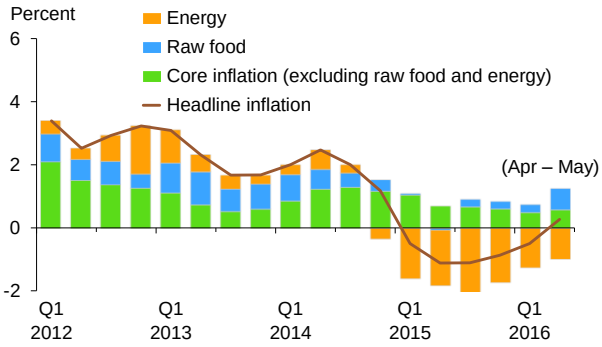
2.3 Production cost and price conditions

Headline inflation turned positive thanks to an increase in fresh food and oil prices, while core inflation edged up slightly on account of a modest increase in costs and demand pressures following the slow pace of domestic demand recovery. The Committee anticipated a gradual rise in headline inflation on the back of rising global oil prices, and would continue to closely monitor domestic demand recovery going forward.

Headline inflation turned positive in the second quarter of 2016, with the average for the first two months of 0.26 percent (yoy) **(Chart 2.14)**, on account of rising fresh food and retail oil prices. The increase in fresh food prices was due to a decline in the vegetables and fruits supply as a result of the drought. However, fresh food prices were expected to drop once rainfall returned to normal. Meanwhile, domestic retail oil prices trended up in line with global crude prices following the decline in US and OPEC oil supply and the slight increase in oil demand, in particular from China and India.

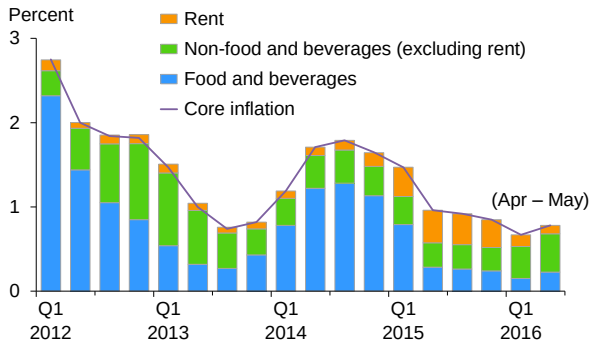
Core inflation for the first two months of the second quarter averaged at 0.78 percent (yoy) **(Chart 2.15)**, up slightly from the last quarter, as food and beverage prices rose following an

Chart 2.14 Contribution to headline inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce calculations by Bank of Thailand

Chart 2.15 Contributions to core inflation

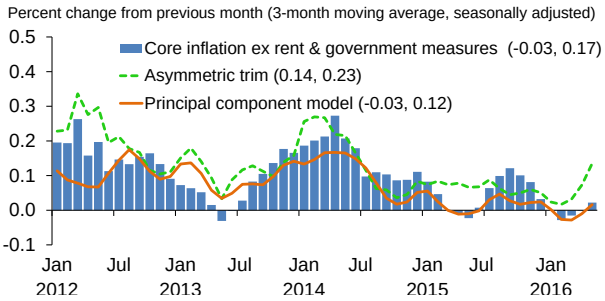


Source: Bureau of Trade and Economic Indices, Ministry of Commerce calculations by Bank of Thailand

increase in fresh food prices. This was consistent with a finding that one percentage increase in fresh food prices would raise core inflation by 0.02 percent [1/]. At the same time the recovery of domestic demand induced more demand-pull inflationary pressure, as reflected in a slightly upward trend of underlying inflation indicators **(Chart 2.16)**. Nonetheless, most prices of goods and services in the core consumer price basket remained largely unchanged **(Chart 2.17),** reflecting limited price adjustments. Short-term inflation expectations edged up slightly in line with in rising inflationary pressure. One-year-ahead inflation expectations according to surveys of professional economists and businesses, respectively, stood at 1.9 percent as of June 2016 and 2.03 percent as of May 2016. Medium-term inflation expectations remained near the inflation target, with five-year-ahead inflation expectations based on a survey of professional economists staying at 2.5 percent as of April 2016 **(Chart 2.18)**.

Looking ahead, the Committee estimated that headline inflation would remain positive and gradually rise on the back of an upward trend in global crude prices. In any case, the Committee would continue to monitor the recovery of domestic demand and developments of inflation expectations closely.

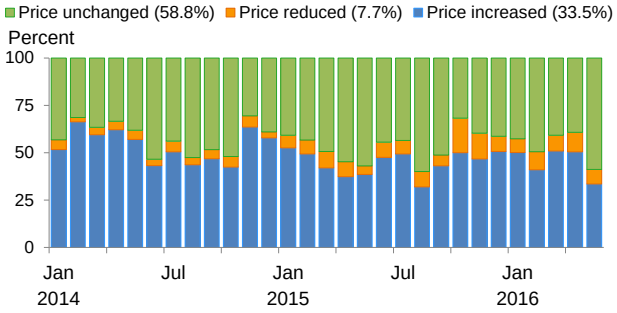
Chart 2.16 Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of May 2016, while the second value is 2004-2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile from the distribution of price changes; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

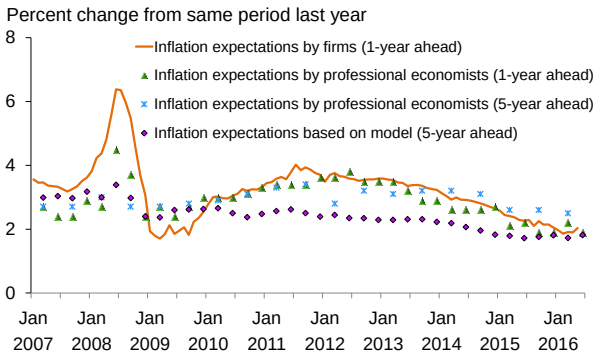
Chart 2.17 Share of items with different price adjustments in the core CPI basket



Note: Price changes calculated from %MoM data
Data in () indicates share of items with that price adjustment in May 2016

Source: Ministry of Commerce, and calculations by Bank of Thailand

Chart 2.18 Inflation expectations



Sources: Business Sentiment Survey of Bank of Thailand, Consensus Forecast, and calculations based on macro-finance term structure model

[1/] The model used for estimating the pass-through of costs to core inflation is given by

$$INF_t = a + bINF_{t-1} + cFarmP_t + dX_t + e_t$$

where INF_t is core inflation, $FarmP_t$ is a fresh food price index, X_t is a set of variables that influence inflation such as the baht per US dollar exchange rate, Dubai crude oil price, and the output gap.

Table 2.1 Quarterly inflation

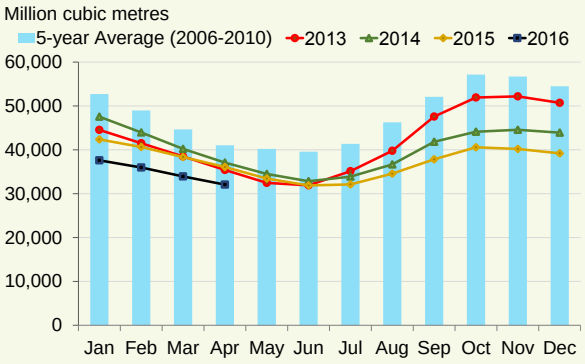
Unit : Percent	2014	2015				2016	
		Q1	Q2	Q3	Q4	Q1	Apr-May
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index	1.89	-0.50	-1.12	-1.11	-0.86	-0.50	0.26
• Core Consumer Price Index	1.59	1.47	0.96	0.92	0.85	0.67	0.78
• Raw food	3.46	0.30	-0.40	1.48	1.45	1.52	3.92
• Energy	1.68	-12.95	-14.16	-16.42	-14.63	-11.41	-8.82
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index	-	-0.7	-0.2	0.0	0.0	-0.3	-
• Core Consumer Price Index	-	0.3	0.1	0.3	0.2	0.1	-
• Raw food	-	0.0	-0.3	1.3	0.4	0.0	-
• Energy	-	-8.2	-1.1	-3.9	-2.2	-4.6	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce.
Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

The economic impacts of drought on agriculture at the regional level

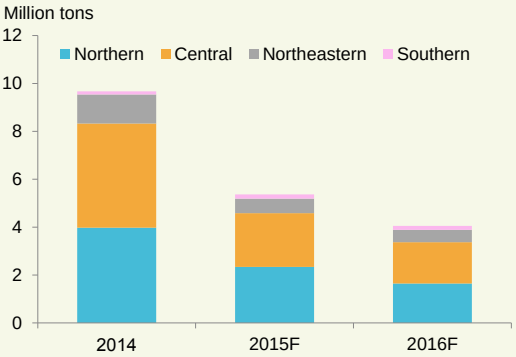
The drought caused by El Niño has impacted Thailand's agricultural sector since the end of 2014. Rainfall fell below average and dam water for irrigation stayed at a low level **(Chart 1)**. The drought adversely affected farmers who depended on water in irrigated areas—estimated to be more than 30 million *rai,* or 20 percent of land countrywide used for agricultural purposes. The most affected area was the Chao Phraya and Mae Klong watersheds where off-season rice was planted. As a result, off-season rice production in 2016 was estimated to decline not only from the previous year's level but much below the level in 2014 that was considered a normal year **(Chart 2)**. In addition, the current drought was widely expected to be more severe than those in the past and would affect almost all croppers nationwide.

Chart 1 Thailand total dam water levels, monthly (2013-2016) and 5-year Average



Source: Royal Irrigation Department

Chart 2 Off-season rice production by region



Note: 2015 and 2016 data are forecasted
Source: Office of Agricultural Economics

The impacts on farm income and responses by farmers

Farmers in each region were affected differently depending on geographical factors and types of plants grown. Northern and central regions, which mainly grow off-season rice, suffered more than other regions. Of the central region, the lower part was most affected as it was at the tail-end and downstream of the irrigation scheme. Meanwhile, the lower part of the northern region and the upper part of the central region, located at the upstream of the irrigation scheme, were second most affected areas. The northeastern region was not much impacted because most farmers did not grow off-season rice. The southern region suffered least because rubber and palm oil, grown by most farmers there, were drought-tolerant relative to rice.

For this year's drought season, most farmers stop growing off-season rice[1/]. Farmers in the central and northern regions reduce cropping areas by 80 and 70 percent, respectively. While only a small number of farmers, whose cropping areas located near water resources, can grow and harvest, their production is estimated to fall by 20 percent relative to a normal year[2/]. As a result, most farmers wait to grow in-season rice. This year's overall production is therefore

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[1/] Based on a survey of 224 farmers from 30 provinces in the central, northern, northeastern, and southern regions in April 2016.

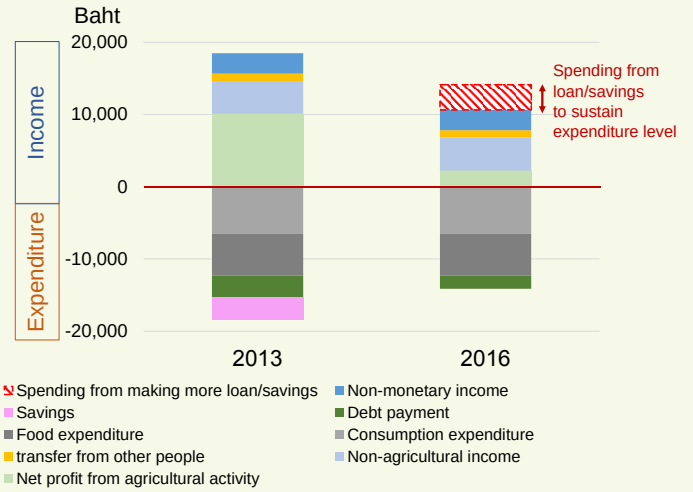
[2/] Off-season rice production in a normal year was on average 1,000 kilograms per rai. In 2016, this drops to 800 kilograms per rai.

projected to drop by 20 percent relative to a normal year[3/]. In addition, with current paddy prices falling by 30 percent from the earlier period, overall farm income is estimated to decrease by 45 percent from the normal level.

Drought-affected farmers have limited options in response. First, most farmers are of old age and thus not flexible in moving out to the non-agricultural sector. Second, farmers cannot resort to growing drought-tolerant plants entirely, as water is still not possibly sufficient and the Royal Irrigation Department no longer distributes water in some areas. Third, the availability of nonfarm jobs, such as those in construction or paid daily works, is limited and varies depending on slack in each locality's industries. Thus, it is not always possible to absorb all workers from the agricultural sector.

Such limited response options cause most farmers to adapt financially by borrowing from banks or loan sharks in order to maintain their subsistent level of consumption **(Chart 3)**. Although some of them can rely on temporary transfers from their children working in the non-agricultural sector, this cannot completely offset the drought impact. Thus, overall purchasing power of the agricultural households declines.

Chart 3 Monthly income-expenditure structure of rice farmer in the central region case study



Source: Household Socio-economic Survey 2013, and calculated by the Bank of Thailand

The sample includes only quartile 1 to 3 of rice farmers in the central region with assumption as follows:

(1) Of-season rice cropping area would drop to 20 percent in 2016
(2) Yield per rai would drop to 80 percent (3) Rice price would drop 30 percent
(4) Sample includes only of-season rice farmers who could grow to pay loan's interest rate.
(5) In-season rice production turned back to a normal level
(6) Income from other sources had not changed
(7) Other expenditure excluding interest rate had not changed

The drought also affects agriculture-related businesses[4/]. More than 50 percent of **rice mills** nationwide have to close their business temporarily, while some that still operate reduce hiring to maintain their financial liquidity[5/]. **Suppliers of farm inputs,** with little adaptation, lose their revenues by 50-80 percent relative to last year, as they decide to just wait for the next growing season. In contrast, **agricultural equipment stores** lose their revenues by only 30-40 percent from last year, as they adapt by launching sales promotion and cutting costs. Some of them resort to selling other machinery and equipment, such as water pumps, that are currently in demand. Overall, income of agriculture-related business owners and workers drops, which will further negatively affect consumption.

[3/] The estimation is based on the assumption that drought impacts only on off-season rice production, while in-season rice production is at a normal level.

[4/] Based on a survey of 19 agricultural machinery and equipment stores as well as rice mills in the central, northern, and northeastern regions.

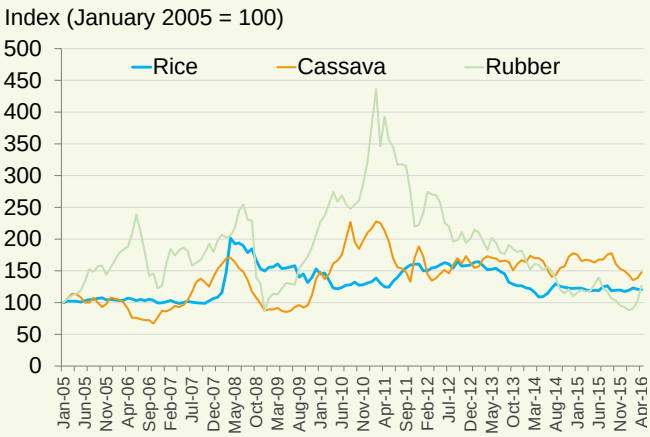
[5/] Thai Rice Mills Association

Such decreasing purchasing power causes consumption in the areas where farming is the key economic activity to slow down. In the provinces where primary customers are farmers, registration of pickup trucks and motorcycles dropped by 27 percent and 7 percent, respectively, in the first quarter of 2016 relative to the normal year averages (2013-2014). In addition, consumer confidence declined to a low level, as reflected in the regional consumer confidence index which stayed below 50 and continued to fall since the beginning of this year[6/]. Although sales of household consumer goods expanded in the first quarter of 2016, that was mainly attributable to sales of necessary goods which still grew at a rate close to a normal year rate, whereas sales of unnecessary goods slowed down from the normal growth rate.

The outlook for farm income and consumption

With the impact of the drought moderating, the Committee expects an improved outlook for farm income and consumption, especially when farmers return to normal activity once rainfall resumes its normal pattern. In addition, agricultural prices began to trend up following the increases in global oil and commodity prices. However, farm income and consumption continue to be weighed on by the following factors. **First, despite rising, agricultural prices will not return to the high levels as observed in the past (Chart 4).** Although rubber prices grow in line with increasing oil prices, together with the international agreement to decrease rubber exports among major producers (i.e., Thailand, Indonesia, and Malaysia), China's low demand will still exert downward pressure on rubber prices. Similarly, prices of Thai rice will continue to be held down by large stocks. **Second, debt burden of farmers remains high**, in part because of their borrowing to sustain necessary consumption since the drought which started at the end of 2014. Furthermore, the level of debt is likely to trend up, as farmers need to borrow more for their consumption and farming activities prior to obtaining their earnings from the upcoming harvest.

Chart 4 Agricultural Price Index



Source: Office of Agricultural Economics and calculated by the Bank of Thailand

In summary, although farmers were able to start growing at the end of the second quarter and are expected to harvest in the fourth quarter, which is in line with the normal pattern, low farm prices and debt repayment will cause farmers' consumption to remain at a low level for a while before returning to normal. The Committee will continue to monitor these developments closely.

[6/] Based on the survey by the Bureau of Trade and Economic Indices of the Ministry of Commerce. The index that is below 50 indicates a drop of consumer confidence from the previous month (i.e., the number of respondents with greater confidence in economic conditions is smaller than those reporting less confidence).

Monetary and Financial Stability

3. Monetary and Financial Stability

Volatility in global financial markets subsided at the beginning of the second quarter of 2016, compared with early in the year, thanks to reduced concerns over China's financial market risks. But toward the end of the quarter, volatility picked up sharply as a result of investors' portfolio adjustments following the Brexit referendum results. On the domestic front, monetary conditions remained accommodative, with money market rates at low levels. Businesses' costs of debt financing remained low overall, despite some increase relative to the previous quarter. Meanwhile, baht movements were primarily driven by external influences.

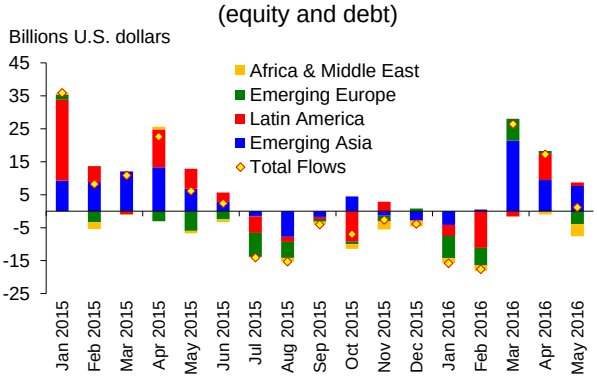
Overall macro-financial stability was not deemed to be a concern, as reflected in commercial banks' high levels of capital buffers and loan loss provisions, as well as the corporate sector's financial strength especially for large corporations. However, households and corporations' debt serviceability declined. As a result, the loan quality of the banking sector deteriorated, particularly for agricultural households, low-income households, as well as small- and medium-sized enterprises (SMEs). Amid the prolonged low interest rate environment, the search-for-yield behavior persisted and seemed poised to increase going forward, although there were limited signs of underpricing of risks at the moment. In addition, the government's real estate stimulus measures, which ended in April 2016, played a key role in expediting customers' purchases. This, coupled with the reduced number of new projects launched by developers, helped mitigate excess supply in the property market. Meanwhile, the fiscal position remained sound.

3.1 Financial markets

Global financial markets

Market volatility subsided at the beginning of the second quarter of 2016, compared with early in the year, as investors' sentiments improved because of reduced concerns over China's financial market risks. Investment in emerging market securities picked up following monetary policy easing in G3 economies, before declining in May 2016 **(Chart 3.1)**. The slowdown was mainly driven

Chart 3.1 Net portfolio flows into Emerging Markets (equity and debt)

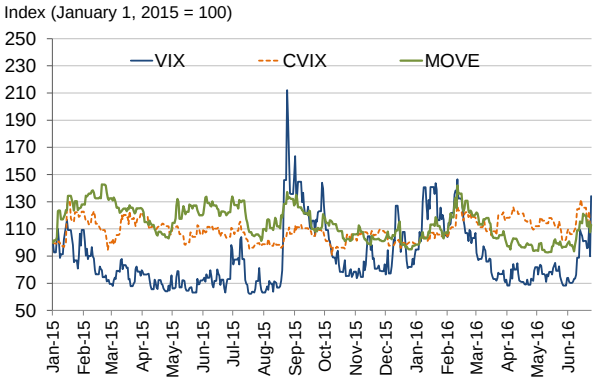


Source: IIF Monthly EM Portfolio Flows Database

by market expectations that the federal funds rate would be raised in the June 2016 meeting of the Federal Open Market Committee (FOMC). Major factors fueling the expectations were (1) the minutes of the April 2016[1/] FOMC meeting, which expressed confidence in the U.S. economic recovery and deemed a policy rate increase as appropriate, and (2) U.S. economic data releases, such as retail sales and inflation, which turned out better than anticipated. Subsequently, market sentiments reversed after the release of disappointing non-farm payroll employment for May 2016[2/] and comments from some FOMC members favoring a gradual normalization path. Expectations of a more delayed rate hike prompted capital flows to return to emerging markets in June 2016.

Toward the end of the second quarter, the United Kingdom's referendum on European Union membership was held on June 23, 2016, and the majority voted in favor of leaving the European Union (Brexit). Following the referendum results, volatility in financial markets surged sharply as indicated by volatility indices[3/] **(Chart 3.2)**. Investors became more concerned and adjusted portfolios in response to changing circumstances. However, the heightened volatility in asset prices and capital

Chart 3.2 Volatility indices



Source: Bloomberg and calculations by Bank of Thailand.

[1/] Minutes of the Federal Open Market Committee (FOMC) Meeting on April 26-27, 2016, released on May 18, 2016.

[2/] The non-farm payroll employment figure for May 2016 was released on June 3, 2016.

[3/] **VIX** (Volatility Index) is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P 500.

MOVE (Merrill Lynch Option Volatility Estimate) is an indicator for bond market volatility, measured by implied volatility of the U.S. government bond market.

CVIX (Currency Volatility Index) is an indicator for foreign exchange market volatility, measured by implied volatility of 9 major currency pairs.

flows was assessed to be transitory and partly anticipated. With regard to the impact of Brexit on the Thai economy, the near-term impact was likely to be limited, given that most sectors were prepared, especially financial institutions whose foreign exchange positions had been hedged. In addition, Thai financial institutions had a limited direct exposure to the UK and the EU, with assets in these countries accounting for only 1.3 percent of total assets. Furthermore, Thailand's external stability remained sound, with international reserves at 3.4 times of short-term external debt, which was assessed to be sufficient to provide cushion for the impact of Brexit.

With regard to the repercussions of Brexit on the Thai economy in the period ahead, there could be some capital outflows from emerging markets, although the impact would be limited given foreign investors' currently reduced exposure in Thai financial markets. Nonetheless, the impact from changes in trade and investment plans by businesses in the UK and the EU, coupled with potential exits by other EU members, could lead to more uncertainties over the periods ahead. The Committee would thus remain vigilant and monitor future developments closely.

Domestic financial markets

Short-term money market rates stayed below the policy rate, while government bond yields rose from the first quarter of 2016 in almost all maturities, thanks to improved confidence of both domestic and foreign investors. However, overall bond yields stayed at low levels and edged lower in early June 2016, following market expectations that a policy rate hike by the Federal Reserve would be delayed.

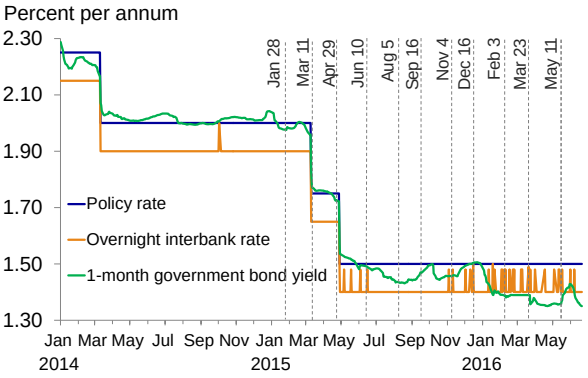
Money, bond, and stock markets

Short-term interest rates stayed below the policy interest rate between April and June 2016 **(Chart 3.3)**. Meanwhile, government bond yields remained at low levels, although higher than the levels seen at the end of the first quarter **(Chart 3.4)**. This was driven mainly by the anticipation of a rate hike by the Federal Reserve in June 2016, following its positive assessment on economic recovery in the US. Furthermore, domestic sentiments improved following the release of Thailand's GDP for the first quarter of 2016 that turned out better than expected. However, Thai government bond yields edged lower at the beginning of June 2016, after the US non-farm payroll employment figures turned out lower than market expectations, causing investors to anticipate a delayed policy rate hike by the Federal Reserve.

Credit spreads between corporate and government bond yields were largely stable relative to the previous quarter. As a result, businesses' costs of financing through corporate bonds rose along with the increase in government bond yields which were used as a benchmark. However, overall financing costs remained below last year's averages **(Chart 3.5)**.

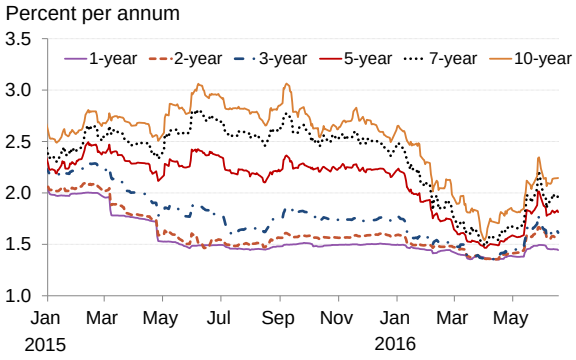
The Stock Exchange of Thailand (SET) Index in the second quarter of 2016 edged up slightly from the previous quarter (Chart 3.6). Major supporting factors were as follows. First, foreign investors started to return to the domestic equity market and build up positions gradually, thanks in part to the relatively solid performance of listed companies compared with those in regional countries, coupled with their diversified sources of revenues across domestic and global markets. Second, the prices of large-capitalization stocks,

Chart 3.3 Money market interest rates



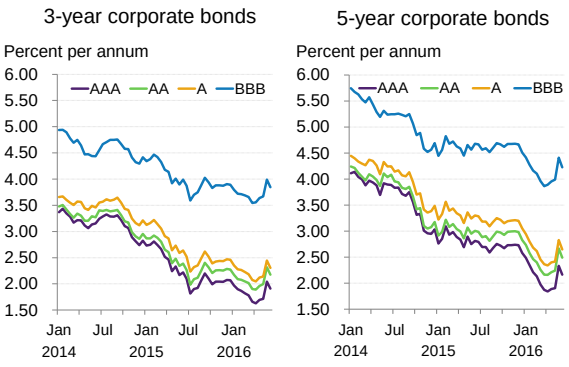
Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 3.4 Thai government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

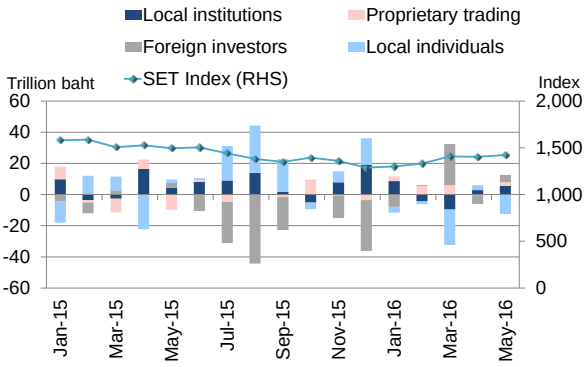
Chart 3.5 Thai corporate bond yields



Source: Thai Bond Market Association (Thai BMA)

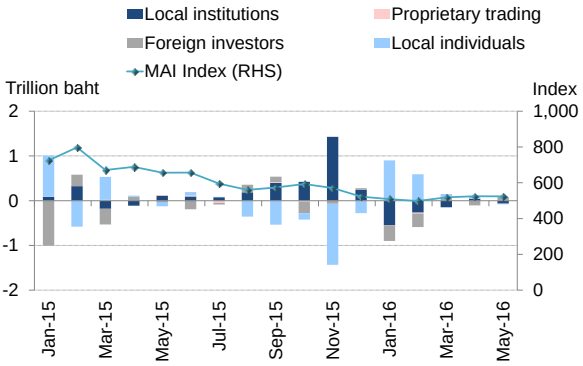
such as those in the energy and wholesale-retail sectors, began to rebound following substantial declines in the earlier periods. Third, two Thai securities were added to the Global MSCI Index in May 2016, which increased foreign investors' demand for Thai equities. Meanwhile, the Market for Alternative Investment (MAI) Index stabilized from the previous quarter **(Chart 3.7)**. Although retail investors recorded net purchases overall, domestic institutional investors had been reducing their investment since the beginning of the year.

Chart 3.6 Stock Exchange of Thailand (SET) Index and net buy classified by investor types



Source: Stock Exchange of Thailand

Chart 3.7 Market for Alternative Investment (MAI) Index and net buy classified by investor types



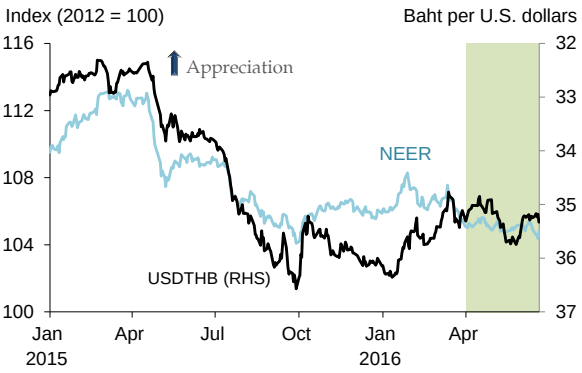
Source: Stock Exchange of Thailand

Foreign exchange markets

Movements of regional currencies were largely driven by investors' expectations about the timing of a policy rate hike by the Federal Reserve and the Brexit referendum results. Relative to the end of the first quarter, the baht depreciated against the US dollar in the second quarter of 2016, in line with movements of most regional currencies. This was particularly the case for May, during which investors widely anticipated a policy rate hike by the Federal Reserve in June 2016. Moreover, the baht also weakened further following the release of the press statement of the MPC meeting on May 11, 2016, which expressed a concern that the strength of the baht relative to certain trading partners' currencies might not be conducive for economic recovery.

Nonetheless, the baht and other regional currencies then appreciated at the beginning of June 2016, given the anticipation of a delayed rate hike by the Federal Reserve, which followed the release of the US non-farm payroll employment figures that turned out lower than market expectations. This prompted some investors to return to regional markets, strengthening the baht and regional currencies. Some correction was then observed after the results of the Brexit referendum. This led investors to increase holdings in safe-haven

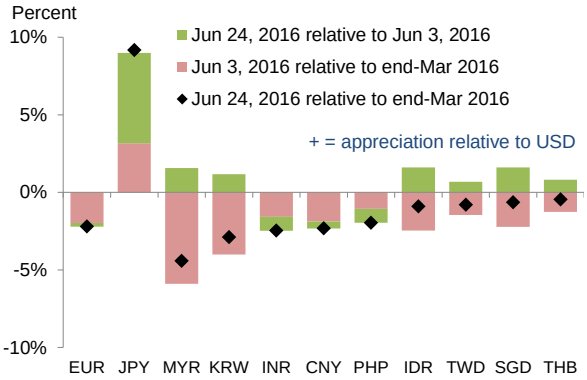
Chart 3.8 Movements in USDTHB and NEER



Source: Bank of Thailand and Reuters (latest data on June 24, 2016)

assets, such as dollar- and yen-denominated assets as well as gold. As of market close on June 24, 2016, the baht stood at 35.33 per US dollar, depreciating by 0.5 percent from the end of the previous quarter. This was in line with the Nominal Effective Exchange Rate (NEER) that closed at 104.80, depreciating by 0.6 percent from the end of the previous quarter mainly due to baht's relative weakness against the Japanese yen **(Chart 3.8 and 3.9)**.

Chart 3.9 Currency movements vis-à-vis USD



Source: Bank of Thailand and Reuters (latest data on June 24, 2016)

3.2 Financial institutions

Private-sector costs of financing through commercial banks edged lower and deposit rates declined marginally. Meanwhile, the private sector's overall demand for new financing slowed from the fourth quarter of 2015, both from new credits and debt financing, in line with the gradual economic recovery. The outlook of a slowdown in credit growth also reduced the need for deposit mobilization, as reflected in the decline in issuance of special deposit products.

Interest rates, credits, and deposits

Costs of funds raised from commercial banks declined in the second quarter of 2016, as large commercial banks reduced their loan interest rates in early April 2016, which prompted medium and small banks to follow suit as well **(Table 3.1)**. This pointed to ample liquidity accumulated in the commercial bank system, partly attributable to slowing growth of commercial bank credits. While the reduction in loan rates could relax monetary conditions further and ease the burden of indebted businesses to some extent, the benefit on private-sector demand for new credits was not likely to be significant given low investment demand overall.

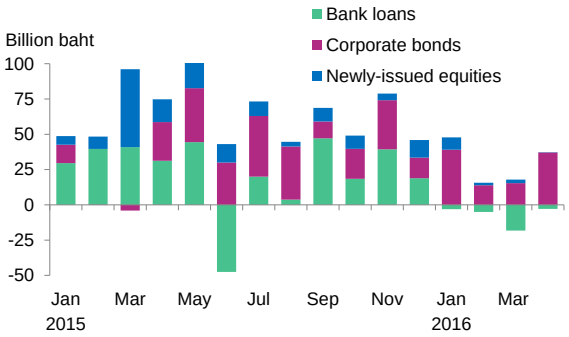
Table 3.1 Benchmark loans and deposits rates of commercial banks*

Percent	2014	2015			2016	
		Q2	Q3	Q4	Q1	May 19
12-month deposits						
Average of the 4 largest Commercal banks**	1.73	1.50	1.46	1.40	1.40	1.38
Average of remaining banks***	2.22	1.75	1.72	1.59	1.47	1.42
Minimum lending rate (MLR)						
Average of the 4 largest commercial banks	6.75	6.51	6.51	6.51	6.51	6.26
Average of remaining banks	7.44	7.32	7.31	7.25	7.25	7.16
Minimum retail rate (MRR)						
Average of the 4 largest commercial banks	8.08	7.86	7.87	7.87	7.87	7.62
Average of remaining banks	8.63	8.54	8.54	8.54	8.54	8.45

Note: * Benchmark rates averaged across commercial banks at the end of each period.
** Four largest commercial banks are BBL, KTB, KBANK and SCB.
*** The following commercial banks are now included under remaining banks: Bank of China (Thailand) (added August 2014), ANZ (Thailand) (added June 2015), and Sumitomo Mitsui Trust Bank (Thailand) (added October 2015). The 12-month average deposit rate of remaining banks in the fourth quarter of 2015 had declined from previous quarter partly because Sumitomo Mitsui Trust Bank (Thailand) offered relatively lower rates.

The amount of total funds raised by the corporate sector between February and April 2016 declined from the fourth quarter of 2015 **(Chart 3.10)**, with some signs of weaknesses in bank loans after the low-interest loan program for SMEs ended in January 2016. The decrease in bank loans was concentrated in the manufacturing and real estate sectors. On the other hand, loan growth was observed in a limited number of industries, namely the electricity production, hotel, and telecommunications industries. Corporate loans were also constrained by strict credit standards imposed by financial institutions, given their concerns over economic conditions and the deteriorated quality of corporate loans. Meanwhile, some businesses appeared to shift their financing from bank loans to corporate bonds, especially those in the construction material, hotel, food products, and construction sectors. However,

Chart 3.10 Total corporate financing by instrument*



Note: *Monthly change in corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.
Source: Bank of Thailand and ThaiBMA

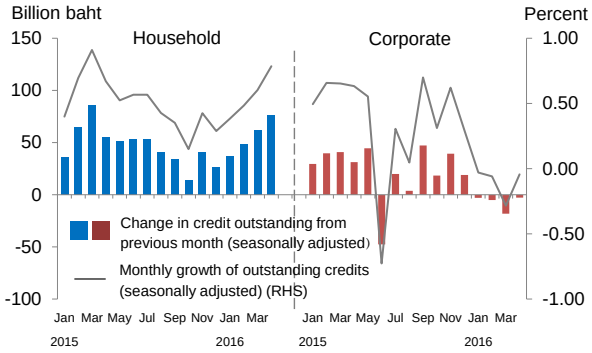
overall funds raised from debt financing remained stable from the fourth quarter of 2015.

New loans to households[4/] picked up between February and April 2016 relative to the fourth quarter of 2015 **(Chart 3.11)**. Part of the increase was supported by temporary factors, including (1) the government's real estate stimulus measures, which ended in April 2016, and (2) the tax-reduction incentives for consumption and tourism during Thailand's New Year festival. In addition, the rise in housing and credit card loans was also supported by seasonal spending patterns. The Credit Conditions Survey for the first quarter of 2016 indicated that financial institutions increased their lending standards for households from the previous quarter due to concerns over economic conditions, borrowers' creditworthiness, as well as the prospect of the housing market.

Deposits outstanding at the end of April 2016 increased from the end of January, attributable to both household and corporate deposits. However, growth of deposits (including bills of exchange, B/E) remained low **(Chart 3.12)**, as the slowdown in both current credit conditions and future trend reduced the need for deposit mobilization. This was reflected in a continued reduction in interest rates on special deposit products **(Chart 3.13)**.

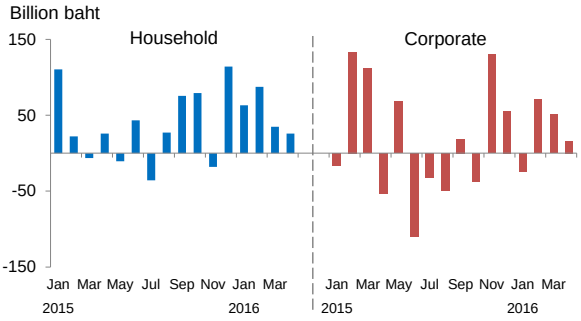
Overall, **the loan-to-deposit (including B/E) ratio** declined from the fourth quarter of 2015 due mainly to slowing growth of new credits **(Chart 3.14)**.

Chart 3.11 New private credits



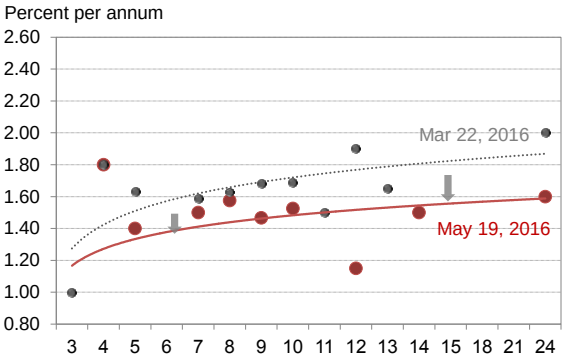
Note: New corporate credits calculated from seasonally adjusted changed in credits outstanding of Other Depository Corporations (ODCs)

Chart 3.12 New deposits*



Note: *Change in outstanding deposits at depository institutions (excluding Bank of Thailand) from the previous month. The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.

Chart 3.13 Special deposit rates*



Note: *Maximum rates offered by commercial banks
Source: Bank of Thailand

Chart 3.14 Loan to deposit (including B/E) ratio of commercial banks



Source: Bank of Thailand

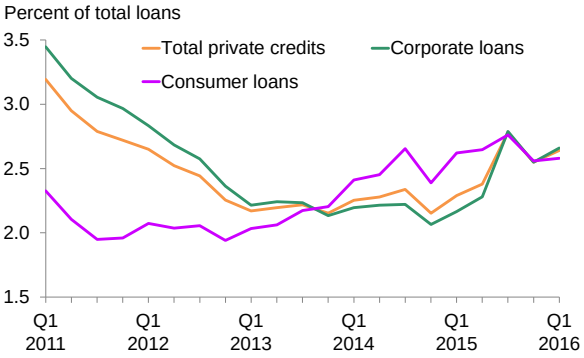
[4/] New loans are calculated from the changes of seasonally adjusted credit outstanding of other depository corporations (ODCs), which include commercial banks, specialized financial institutions, finance companies, savings cooperatives, and money market funds.

Overall macro-financial stability was not deemed to be a concern, as reflected in solid financial positions of financial institutions and large businesses. Nevertheless, loan quality, especially those of SMEs and agricultural households, deteriorated due to both the sluggish economic recovery and the impact of the recent drought. Meanwhile, the stability of financial institutions remained sound. This was because commercial banks continued to monitor credit quality closely and hold high levels of loan loss provisions and capital buffers, which were high enough to provide cushion against risks from the deteriorated loan quality.

Commercial banks' loan quality at the end of the first quarter of 2016 deteriorated from the end of 2015 in line with sluggish economic recovery. This was reflected in the ratio of non-performing loans (NPLs) to total loans that increased from 2.55 percent at the end of 2015 to 2.64 percent **(Chart 3.15)**. The increase in the NPL ratio of corporate loans was attributable to loans extended to SMEs, particularly those in the retail-wholesale sector, which was affected by the slow economic recovery. On the other hand, the NPL ratio of consumption loans edged up slightly due to the deteriorated quality of housing and credit card loans **(Chart 3.16)**.

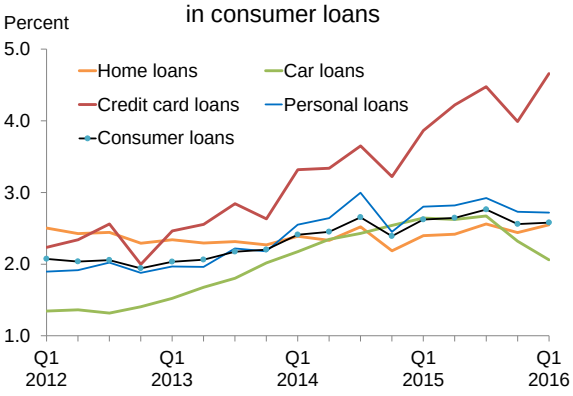
Despite the deterioration in the quality of commercial bank loans, the stability of the commercial bank system remained sound overall. In the recent periods, commercial banks continued to monitor loan quality closely and maintain sufficiently high levels of capital buffers and loan loss provisions, as reflected in the actual/regulatory loan loss provision ratio at 160.0 percent, as well as the capital adequacy ratio at

Chart 3.15 Non-performing loans of commercial banks



Source: Bank of Thailand

Chart 3.16 Share of non-performing loans in consumer loans



Source: Bank of Thailand

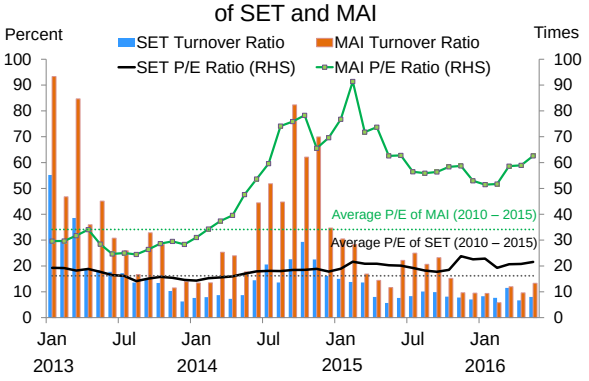
17.5 percent, which was well above the regulatory required level of 8.5 percent. Looking forward, the Committee would continue to monitor the quality of credits that would likely worsen with sluggish economic conditions. This was particularly the case for loans to SMEs and consumption loans, which could weigh on profitability of commercial banks in the period ahead.

The search-for-yield behavior continued to increase under the prolonged low interest rate environment, but there were no broad-based signs of underpricing of risks at the moment. The domestic equity market was still far from overheating, as indicated by the turnover ratio in the SET and the MAI that remained below historical averages. While the price-earnings (P/E) ratios for the MAI continued to stay above historical averages **(Chart 3.17)**, risks to the overall stability of the financial system were assessed to be limited, as these stocks had small market capitalization and investors conducted transactions mostly in cash.

Regarding the bond markets, low levels of bond yields helped attract the corporate sector to rely more on debt financing. Part of this was the issuance of non-rated bonds, with the number of issuing companies growing over time **(Chart 3.18)**. **Despite the recent growth of non-rated bonds, systemic risks were assessed to be limited, as non-rated bonds constituted only a small proportion of total corporate bonds outstanding** and the sales of non-rated bonds were mainly restricted to accredited investors. Nonetheless, it remained necessary to ensure that investors conduct proper risk assessments that were in alignment with issuing companies' financial positions and debt serviceability.

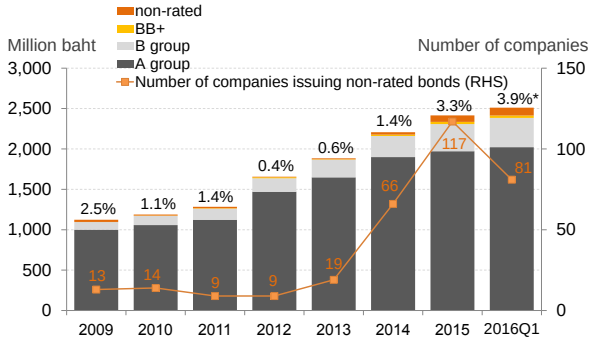
Furthermore, it was also observed that investors increased their holdings of mutual

Chart 3.17 Turnover ratio and Price-Earnings (P/E) ratio* of SET and MAI



Note: * Current price to Earnings Per Share (EPS) over the last 12 months
Sources: Stock Exchange of Thailand and Bloomberg
(latest data on Jun 2, 2016)

Chart 3.18 Corporate bonds outstanding (short and long maturities) classified by credit rating



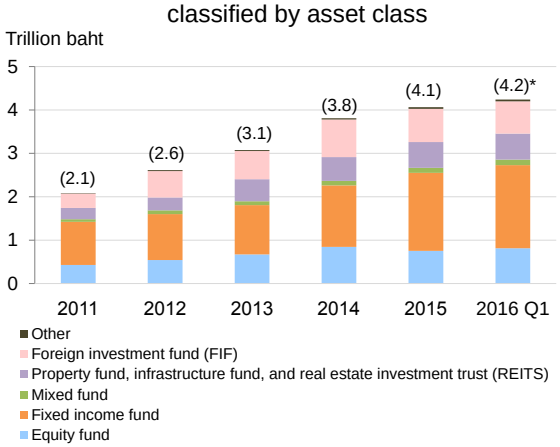
Note: The A group consists of AAA to A-, while the B group consists of BBB+, BBB, and BBB-
* Percent of non-rated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

funds. Considering the net asset value under management of mutual funds in the first quarter of 2016, the amount invested increased in almost all types of mutual funds, with high growth in deposit-like funds such as fixed income funds. Growth was also observed in riskier assets, such as domestic small-capitalization equities and foreign equities. Investment in foreign investment funds (FIFs), in particular, stayed elevated **(Chart 3.19)**. It should be noted, however, that FIFs recently reduced investment in countries with low credit ratings and moved toward higher-rated countries such as China, the United Arab Emirates, and Qatar, with investment mostly in investment-grade bonds.

As an alternative for those seeking yields above bank deposit rates, **savings cooperatives' deposits and shares** started to gain traction among savers, with a sustained growth over the recent periods **(Chart 3.20)**. The main reason was that savings cooperatives' deposit interest rates and dividend yields were higher than those offered by commercial banks. **It was also observed that cooperatives which received additional capitals exhibited the search-for-yield behavior.** For instance, savings cooperatives invested a larger proportion of their assets in debt and equity securities with higher risks and returns, as seen in the pick-up in debt and equity investments **(Chart 3.21)**, though the shares were still small relative to total assets.

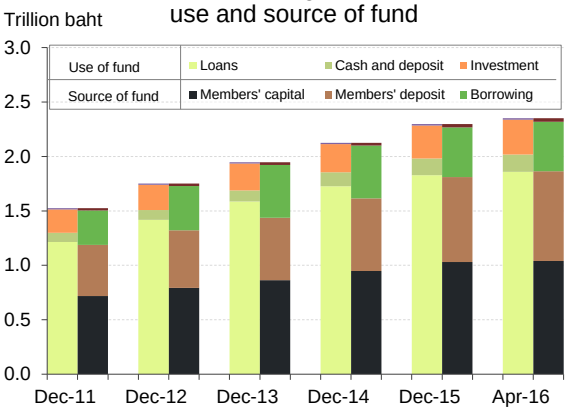
Given the low interest rate environment, investors would likely pursue higher returns by tolerating higher risks. Underpricing of risks remained an issue that warranted close monitoring in the periods ahead. Vulnerabilities in the financial system could accumulate, as a result of both investors' behaviors mentioned above and competition among financial institutions to extend

Chart 3.19 Net asset of mutual funds classified by asset class



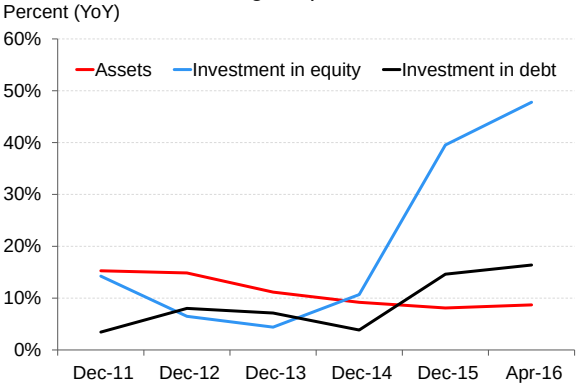
Note: *Data in () indicates total asset of mutual funds
Source: Association of Investment Management Companies, calculations by Bank of Thailand.

Chart 3.20 Saving cooperatives' use and source of fund



Source: Cooperative Auditing Department, calculations by Bank of Thailand

Chart 3.21 Growth in asset and investment of saving cooperatives



Source: Cooperative Auditing Department, calculations by Bank of Thailand

credits to large businesses only to some business sectors. **While broad-based signs of underpricing of risks were not present at the moment, the Committee would continue to monitor the search-for-yield behavior vigilantly going forward.**

3.3 Non-financial sectors

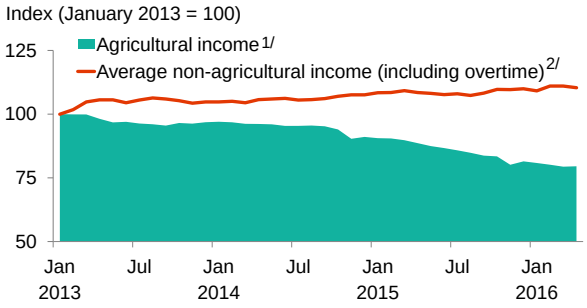
Household sector

Sluggish economic recovery and elevated household debt relative to income continued to weigh on households' debt serviceability. This was particularly the case for agricultural and low-income households, whose liquidity tended to be more constrained than other groups of households.

Sluggish economic recovery continued to weigh on households' income and debt serviceability. Farm income continued to edge lower due to depressed farm prices and the impact from the recent drought, while non-farm income remained rather stable **(Chart 3.22)**. The overall slowdown in household income, coupled with household debt that had surged in the earlier periods, impaired households' debt serviceability. This was reflected in the increase in the NPL ratio of consumer loans, which rose to 2.58 percent at the end of the first quarter of 2016, mainly due to the deteriorated quality of credit card loans.

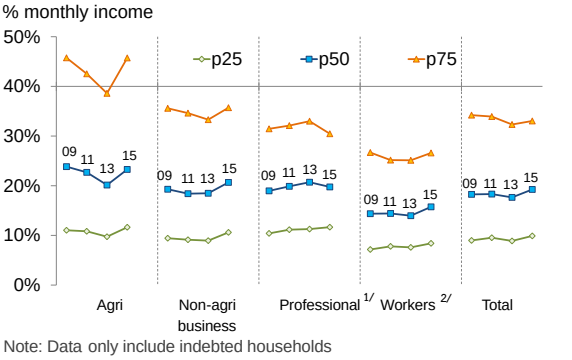
According to the Socio-Economic Survey (SES), agricultural households had higher default risks compared with households in other sectors, as reflected in the debt service ratio (DSR) that stayed elevated **(Chart 3.23)**. In addition to this, households with income in the lowest quintile and those with income in the highest quintile had debt-to-annual income ratios that were comparable.

Chart 3.22 Household income



Note: 1/ Seasonally adjusted, 12-month moving average
2/ Seasonally adjusted, 3-month moving average
Sources: Office of Agricultural Economics, National Statistical Office; calculations by Bank of Thailand

Chart 3.23 Debt Service Ratio (DSR)

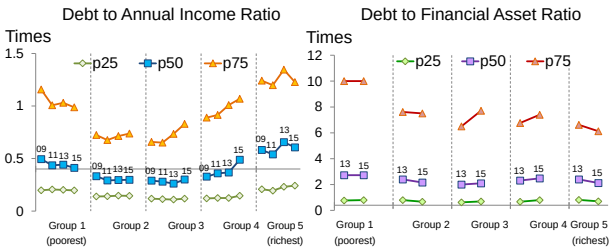


Note: Data only include indebted households
1/ Professionals such as managers, academics, technicians
2/ Workers such as laborers in farming, foresty, fishery ; machine operators, secretaries, waiters, and mechanics
Source: Household Socio-Economic Survey by the National Statistical Office, calculations by Bank of Thailand

However, further inspection of the debt-to-financial asset ratio, which reflected both households' debt and liquidity, suggested the need to monitor low-income households closely, given their high debt and relatively low liquidity compared with other groups of households **(Chart 3.24)**.

Going forward, key developments to be monitored included (1) the outlook for economic recovery, (2) the impact of government stimulus measures, and (3) planting and harvesting after the improved rainfall. It remained to be seen how these factors would support households' income, liquidity, debt serviceability, and consumption over the period ahead, against the backdrop of elevated household debt that further increased over the period of the recent drought.

Chart 3.24 Household debt by income group



Note: Data only includes indebted household

1/ Each income group is defined as follows (based on 2015 data):

Group 1 – Average salary less than 10,000 baht per month

Group 2 – Average salary between 10,000 to 15,000 baht per month

Group 3 – Average salary between 15,000 to 23,000 baht per month

Group 4 – Average salary between 23,000 to 36,000 baht per month

Group 5 – Average salary more than 36,000 baht per month

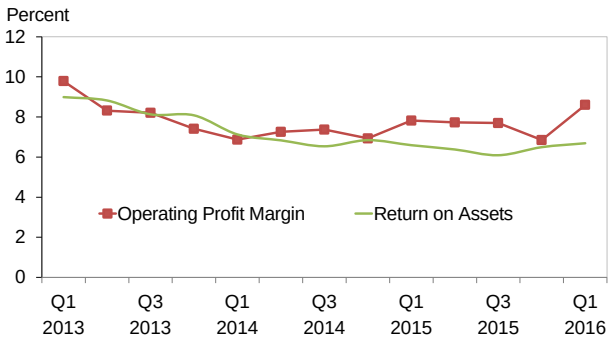
Source: Household Socio-Economic Survey by the National Statistical Office, calculations by the Bank of Thailand

Business sector

Performance of non-financial companies listed in the SET in the first quarter of 2016 was broadly unchanged from the previous year. However, debt serviceability of certain groups of businesses facing liquidity constraints, especially SMEs, could be affected further by the slow economic recovery and thus needed to be monitored in the period ahead.

Overall performance of businesses was broadly unchanged compared with the same period of the previous year, but improved from the previous quarter. Businesses' ability to generate revenues and profits improved, as reflected in the operating profit margin (OPM) and the return on assets (ROA) of most businesses that rose slightly from the previous year **(Chart 3.25)**. The increased profits in some sectors resulted from effective cost management. For instance, businesses in the petroleum exploration and production industry, which had suffered from the decline in crude oil prices in earlier periods,

Chart 3.25 Operating profit margin and return on assets



Note: median estimates

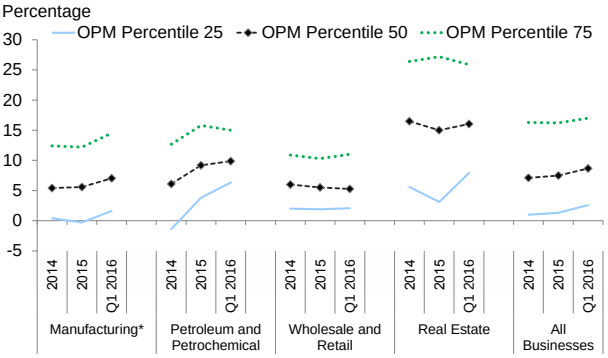
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

appeared to manage their costs more effectively, which translated into lower stock losses. However, some businesses had operating profit margins that were rather stable **(Chart 3.26)**. The real estate sector, for instance, saw improved sales on the back of the government's stimulus measures, even though their profit margins remained stable due to higher costs of land and promotion expenses.

Debt serviceability was healthy overall. Most businesses had adjusted by managing operating costs more effectively, which improved the overall direction of the annualize interest coverage ratio (ICR). Despite a decline in some sectors such as the construction and wholesale-retail sectors, liquidity was ample and the ICR was well above the international standard at 1.5. In addition, business debt increased only slightly, with the median of the debt to equity (D/E) ratio stabilizing at the low level of 0.7.

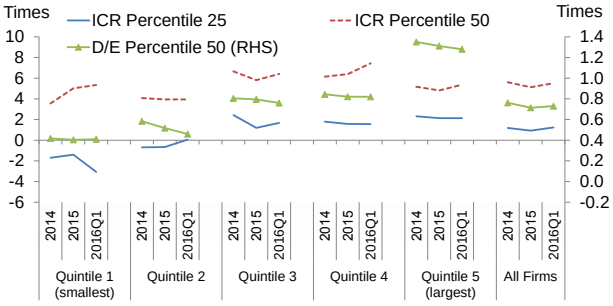
Although businesses' performance and liquidity remained healthy overall, risks in some business sectors continued to warrant monitoring. Examples included businesses that were sensitive to sluggish economic recovery, such as small businesses with limited financing. Incidentally, recent data revealed that listed companies with assets in the lowest two quintiles could be facing liquidity constraints **(Chart 3.27)**. Using these small firms to be representative of SMEs, which were non-listed, it could be implied that debt serviceability of SMEs appeared to deteriorate. This was consistent with the NPL ratio of SMEs at the end of the first quarter of 2016, which was on an upward trend.

Chart 3.26 Operating profit margin by sector



Note: * Exclude Petroleum and Petrochemical
Source: Stock Exchange of Thailand; calculation by Bank of Thailand

Chart 3.27 Interest Coverage Ratio (ICR)* and Debt to Equity (D/E) ratios classified by firm size



Note: *The 25th percentile reflects firms with low debt serviceability; ICR calculation is annualized; D/E is median estimate; Firms divided based on their asset size into 5 quintiles from smallest to largest.
Source: Stock Exchange of Thailand; calculation by Bank of Thailand

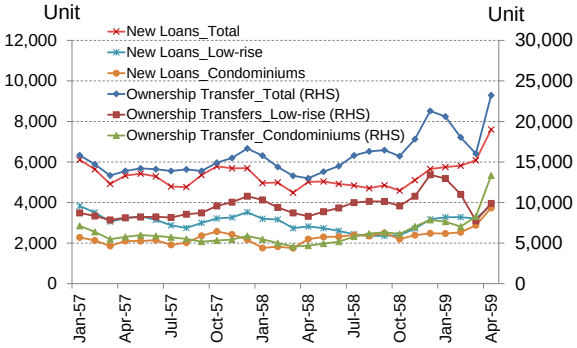
Real estate sector

The property market in Bangkok and its vicinities improved in the first quarter of 2016 on the back of the government's real estate measures. Meanwhile, the slowdown in the number of new projects launched would help mitigate risks of oversupply in the period ahead.

The property market in Bangkok and its vicinities improved in the first quarter of 2016 on the back of the government's real estate stimulus measures, which boosted new purchases prior to the expiration of the measures in April 2016. This was reflected in the number of ownership transfers and new housing loans that picked up sharply **(Chart 3.28)**. From the beginning of the year, real estate developers had also adjusted by launching fewer new projects and focusing instead on selling outstanding units **(Chart 3.29)**. Altogether, both demand and supply factors mentioned above helped reduce excess supply to some degree. It should be noted, however, that the real estate measures prompted consumers to hasten their future purchases. Hence, a possible slowdown in future demand would be an issue to be monitored going forward.

With low interest rates and ample liquidity in the financial system, **real estate developers seemed poised to shift from bank loans to other means of financing, such as corporate bonds, joint ventures, and property funds.** In particular, the share of non-rated bonds to the total amount outstanding had been on the rise **(Chart 3.30)**. It would be essential for investors to follow market developments and have a proper understanding of risks associated with these new forms of investment.

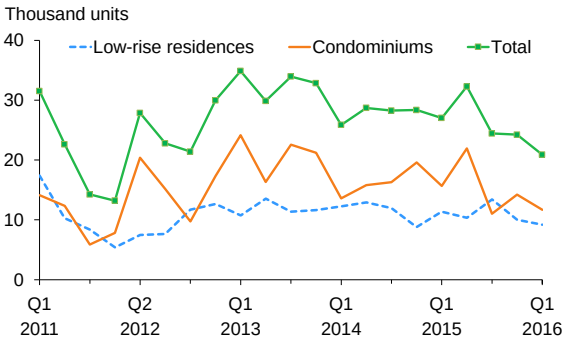
Chart 3.28 Number of new housing loans and ownership transfers in the Bangkok Metropolitan area



Note: Low-rise residences comprised of single houses, semi-detached houses, townhouses, and commercial building

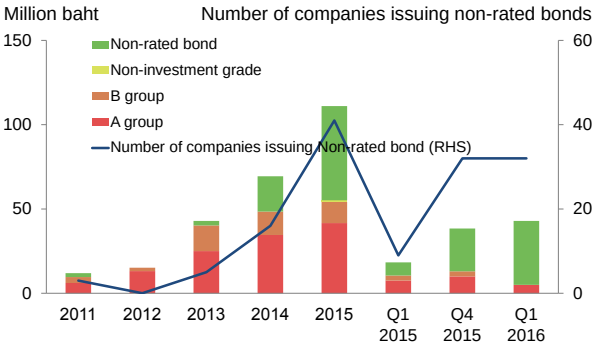
Source: Bank of Thailand and REIC

Chart 3.29 New residential project launched in the Bangkok Metropolitan area



Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

Chart 3.30 Corporate bonds outstanding (short and long maturities)



Note: A group consists of AAA to A-; B group consists of BBB+, BBB, and BBB-; Non-investment grade is BB+

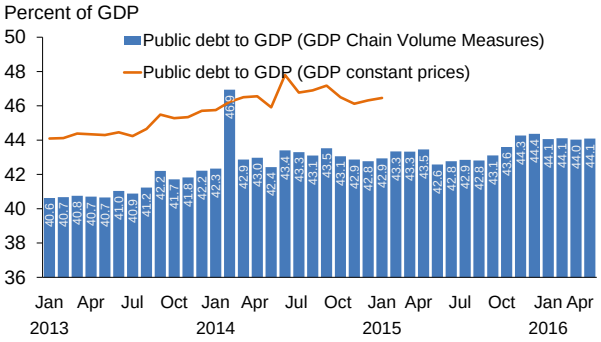
Source: Thai Bond Market Association (Thai BMA)

Fiscal sector

Stability of the fiscal sector was sound, as reflected in the public debt to GDP ratio that remained below the fiscal sustainability threshold and the treasury cash balance that was sufficient to meet unexpected fiscal needs. Nonetheless, pressures on fiscal stability would likely increase from the additional borrowing needed to cover larger budget deficit, the borrowing for investment in transportation infrastructure projects, as well as the future fiscal burden arising from new stimulus measures that had been approved recently.

Stability of the fiscal sector was sound, as reflected in the public debt to GDP ratio that remained below the fiscal sustainability threshold of 60 percent. As of April 2016, the ratio stood at 44.1 percent, down slightly from 44.4 percent in December 2015 **(Chart 3.31)**. The decline was due to (1) the redemption of matured bonds by the Electricity Generating Authority of Thailand, (2) the repayment of foreign debt and the redemption of promissory notes by the Thai Airways International Public Company Limited, and (3) the principal repayment by the Bank of Agriculture and Agricultural Cooperatives. On the other hand, government borrowing increased due to the borrowing to cover budget deficit and the borrowing to conduct debt restructuring for Thai Khem Khang saving bonds, which were scheduled to mature on June 9, 2016. Meanwhile, the treasury cash balance decreased from the end of the previous quarter but remained sufficient to meet unexpected fiscal needs. As of the end of April 2016, the treasury cash balance stood at 206 billion baht, down from 386 billion baht at the end of December 2015. The decline in the treasury cash balance was due to the increase in regular payments resulting from the salary adjustment for government officials,

Chart 3.31 Public debt to GDP ratio*



Note: (1) Chart shows calendar years, (2) Official figures for debt-to-GDP ratio based on GDP chain volume measures are available only for February and March 2015. Figures prior to this period are calculated by the Bank of Thailand

Source: Public Debt Management Office

as well as the continued growth in investment expenditure under several economic stimulus measures.

Over the period ahead, pressures on fiscal stability would likely increase due to (1) the additional borrowing to cover larger budget deficit, (2) the off-budget borrowing for investment in the transportation infrastructure projects under the Urgent Transportation Action Plan, and (3) the additional fiscal burden from quasi-fiscal activities that were part of economic stimulus packages. These quasi-fiscal obligations included (1) the compensation for the reduction in interest income to specialized financial institutions (SFIs) taking part in the soft-loan programs and (2) the compensation given to the Thai Credit Guarantee Corporation for loan guarantee fees for SMEs under the Portfolio Guarantee Scheme 5 (PGS-5) and for loan guarantee fees under the second phase of the guarantee scheme for micro-entrepreneurs.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016		
			H1	Q3	Q4	Q1	Apr	May
1. Financial market sector								
Bond market								
Bond spread								
(10 years-2 years)	1.3	1.1	0.9	1.2	1.1	0.7	0.3	0.5
Equity								
SET Index (End of period)	1,497.7	1,288.0	1,504.6	1,349.0	1,288.0	1,407.7	1,404.6	1,424.3
Actual volatility (SET Index)[1/]	12.5	13.9	12.0	12.3	13.7	15.9	14.3	9.9
Price to Earnings Ratio (times)	17.8	22.6	20.1	17.8	22.6	20.7	20.8	21.6
FX market								
Actual volatility (บาท) (%annualize)[2/]	4.0	5.1	4.5	5.2	6.1	4.4	4.8	4.6
Nominal Effective Exchange Rate (NEER)	104.3	108.5	110.8	106.6	105.9	106.6	105.3	105.0
Real Effective Exchange Rate (REER)	103.0	104.4	106.99	102.4	101.2	100.8	100.0	99.7
2. Financial institutions sector[3/]								
Minimum lending rate (MLR)4/	6.75	6.5	6.51	6.50	6.50	6.51	6.26	6.26
12-month fixed deposit rate4/	1.73	1.40	1.50	1.46	1.40	1.40	1.38	1.38
Capital adequacy								
Regulatory capital to risk-weighted asset (%)	16.8	17.4	16.8	17.3	17.4	17.5	N.A.	N.A.
Earning and profitability								
Net profit (billion baht)	214.1	192.3	103.7	50.0	38.6	51.3	N.A.	N.A.
Return on assets (ROA)	1.2	0.9	1.3	1.2	0.9	1.2	N.A.	N.A.
Liquidity								
Loan to deposit and B/E	95.7	97.0	96.5	97.3	97.0	94.8	N.A.	N.A.
3. Household sector								
Household debt to GDP (times)	79.9	81.5	80.6	80.8	81.5	N.A.	N.A.	N.A.
Financial assets to debt (times)	2.5	2.5	2.5	2.5	2.5	N.A.	N.A.	N.A.
Proportion of loans that are 3 months delinquent (%)								
Thai commercial banks :								
Consumer loans	2.4	2.6	2.6	2.8	2.6	2.6	N.A.	N.A.
Housing loans	2.2	2.4	2.4	2.6	2.4	2.6	N.A.	N.A.
Auto leasing	2.5	2.3	2.6	2.7	2.3	2.1	N.A.	N.A.
Credit cards	3.2	4.0	4.2	4.5	4.0	4.7	N.A.	N.A.
Other personal loans	2.5	2.7	2.8	2.9	2.7	2.7	N.A.	N.A.
4. Non-financial corporate sector[5/]								
Operating profit margin (%)	7.1	7.5	7.8	7.7	6.9	8.7	N.A.	N.A.
Debt to equity ratio (times)	0.8	0.7	0.7	0.7	0.7	0.7	N.A.	N.A.
Income coverage ratio (times)	6.0	5.1	5.2	4.8	5.4	5.5	N.A.	N.A.
Current ratio (times)	1.6	1.7	1.7	1.6	1.7	1.7	N.A.	N.A.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016		
			H1	Q3	Q4	Q1	Apr	May
4. Non-financial corporate sector[5/] (Continues)								
Delinquency and NPL ratio								
Large business	1.1	1.6	2.0	2.0	1.6	1.7	N.A.	N.A.
SMEs	3.1	3.5	3.4	3.6	3.5	3.7	N.A.	N.A.
5. Real estate sector								
Demand								
The number of approved mortgages from banks								
(Bangkok and its vicinity)	62,839	59,667	26,503	15,770	17,394	15,487	6,927	3048.0
Single-detached and semi-detached houses	15,694	13,152	6,384	3,115	3,653	3,511	1,549	6,095
Townhouses and commercial buildings	21,764	19,210	9,199	4,812	5,199	5,092	2,176	1,113
Condominiums	25,381	27,305	10,920	7,843	8,542	6,884	3,202	1,241
Supply								
The number of new openings (Bangkok and its vicinity)	111,211	107,988	59,316	24,436	24,236	20,886	2,905	6,095
Single-detached and semi-detached houses	18,933	17,637	6,354	7,388	3,895	4,178	259	1,040
Townhouses and commercial buildings	26,980	27,518	15,349	6,034	6,135	5,016	863	1,670
Condominiums	65,298	62,833	37,613	11,014	14,206	11,692	1,783	3,385
Housing price index[6/]								
Single-detached houses (including land)	127.9	129.3	127.7	129.6	129.3	129.4	131.1	133.5
Townhouses (including land)	133.5	137.5	133.7	134.7	137.5	138.8	139.3	139.2
Condominiums	151.1	160.9	152.1	156.4	160.9	160.0	160.8	160.1
Land	145.6	168.8	151.9	157.8	168.8	169.5	176.6	175.2
6. Fiscal sector								
Public debt to GDP (%)	46.3	44.4	42.8	43.1	44.4	44.0	44.1	N.A.

[1/] Annualized standard deviation of return

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median); with revisions to data and methodology.

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